Ekagra Karma ❋ Apara Antara

TRAINING INSTITUTE OF HYPNOTHERAPY PVT. LTD.

A COURSE ON

Integrated Clinical Hypnotherapy (Online)

LEVEL V

Course Duration: 048Days
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PROGRAMME SCHEDULE

DAY 1

SR. NO.	TOPIC	TIME
1.	Registration	09:00 – 09:30 AM
2.	Introduction- Concept of Healing; need and role of complementary therapies	09:30 – 10:00 AM
3.	Exercises	10:00 – 10:30 AM
4.	Tea Break	10:30 – 11:00 AM
5.	Chakras	11:00 – 01:00 PM
6.	Lunch	01:00 – 02:00 PM
7.	Chakras (contd)	02:00 – 03:30 PM
8.	Tea Break	03:30 – 04:00 PM
9.	Group meditations	04:00 – 06:00 PM

Day 2:

Sr. No	Content	Duration
1.	Mind Structure	9:00-9:15 A.M.
2.	Talking to the Sub-Conscious Mind	9:15-10:00 A.M.
3.	Spirit- How and Why	10:00-11:00 A.M.
4.	Tea Break	11:00-11:30 A.M.
5.	Dark Force Entities – Theory	11:30-12:45 P.M.
6.	Lunch	12:45-1:45 P.M.
7.	Etheric Abortions	1:45-2:30 P.M.
8.	Vows, Oaths and Curses	2:30-3:30 P.M.
9.	Tea Break	3:30-4:00 P.M.
10.	Group Energies	4:00-4:30 P.M.
11.	Space Energies	4:30-5:00 P.M.
12.	Implants	5:00-5:30 P.M.
13.	Aura Exploration	5:30-6:00 P.M.

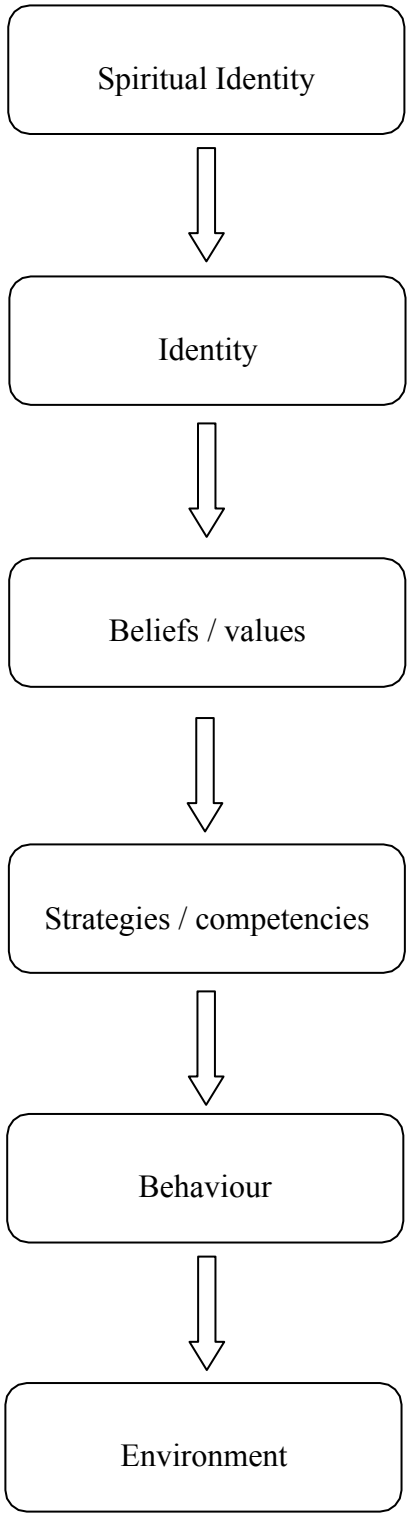
Day 3:

Sr. No	Content	Duration
1.	Journey of Consciousness	9:00-10:45 A.M.
2.	Tea Break	10:45-11:15 A.M.
3.	Metaphysics of disease	11:15-1:00 P.M.
4.	Lunch	01:00- 2:00 P.M.
5.	Metaphysics of disease (continued)	02:00 - 3:30 P.M.
6.	Tea Break	03:30- 4:00P.M.
7.	Metaphysics of disease (continued)	04:00- 06:00 P.M.

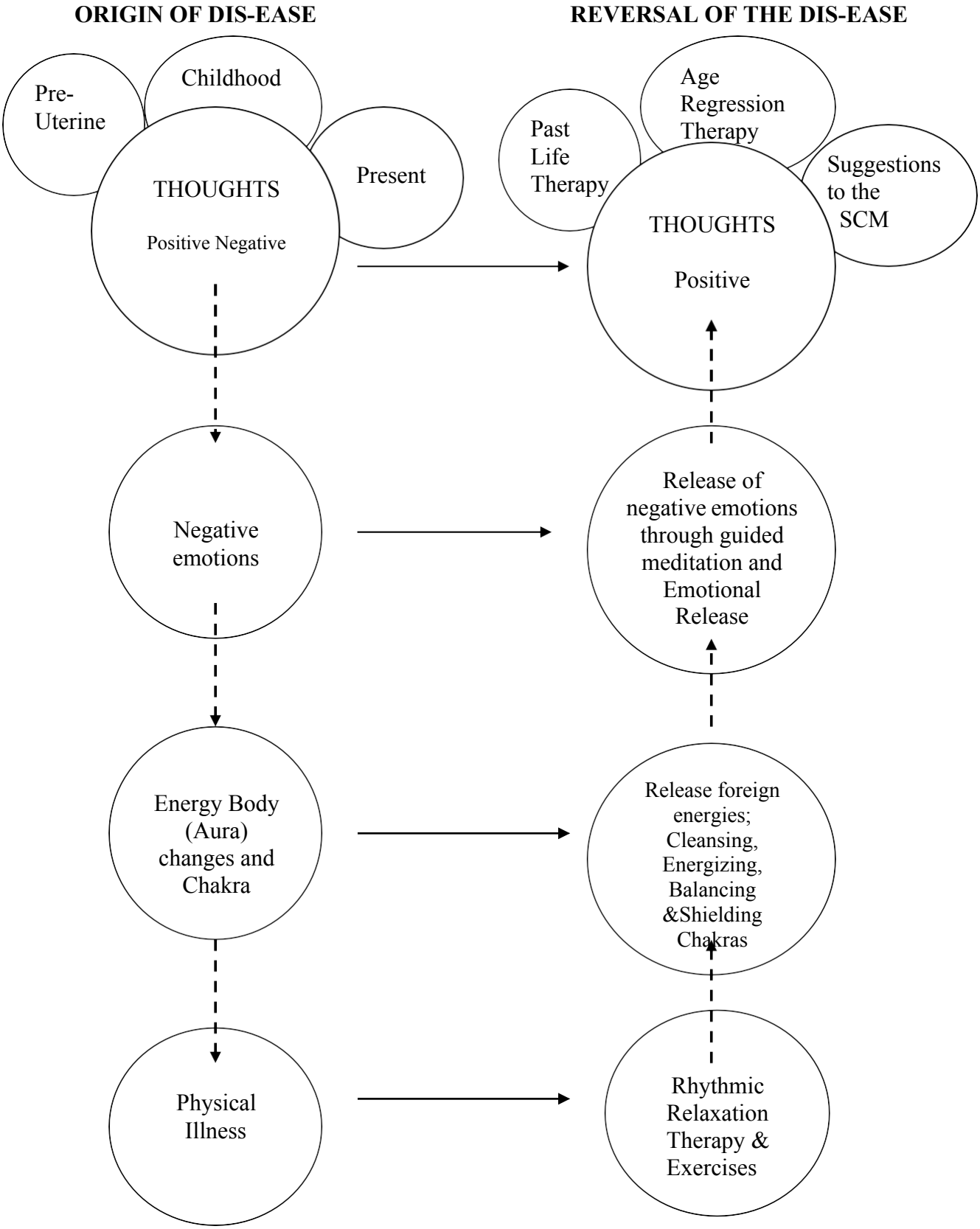
DAY 4:

Sr. No	Content	Duration
1.	Organ Regeneration Theory	09:00 - 10:45 A.M.
2.	Tea Break	10:45 - 11:15 A.M.
3.	Smoking Cessation Theory	11.15 – 12:45 P.M
4.	Lunch	12.45 – 1:45 P.M
5.	Allergy Management Theory	1:45 - 3:00 P.M
6.	Concept Revision	3.00 – 3.30 P.M
7.	Tea Break	3:30 – 4:00 P.M.
8.	Online Test	4:00 – 5:00 P.M.
9.	Feedback, Evaluation & Online Certificate	5:00 – 6:00 P.M.

FLOW OF CONSCIOUSNESS



REVERSIBILITY OF DISEASE



BODY ENERGY

We all search for health and happiness in our external environment, thinking that these are readily available in the outside world. However, several times this quest remains mis-directed. Harmonized functioning of the body is easily achieved by looking inwards and tapping the vast hidden resources of energy present in our system.

Our ancient yogis were living this truth. Most of these techniques were observed by them in nature and further developed to be taught as different forms of yoga postures or asanas.

In the Academy, we have modified some of them to suit modern lifestyle. This helps in:

1. Full utilization of respiratory capacity.
2. Increased oxygen supply to the tissues.
3. Harmonious functioning of the systems.
4. Regulated, proper utilization of energy.
5. Becoming calm and healthy as the mind reaches a silent state.
6. Realization of higher potential leading to greater efficiency.
7. Revitalizing a person to make him fresh, relaxed and dynamic.

BREATHING FOR RELAXATION

Breath is a bridge between Conscious and Sub-conscious Mind, Life and Death, Metaphysical and Physical worlds. When we breathe apart from Oxygen we take in lot of Universal energy.

Breathing is an important function of the body which is under voluntary and in-voluntary control, besides breathing tends to get affected with every thought and emotion. If we can consciously make a difference in our Breathing Patterns we can make a difference in our lives.

1) CLEANSING BREATHING: This pattern is very effective to release any disturbing emotions like anger, stress, frustration etc. Judicious use of this technique restores balance within the human system.

Method: Close your eyes, take a deep breath through the nose with an intention to inhale positive energy and blow out through the mouth with an intension to exhale negative energy or emotions. This cycle may be repeated till the desired state of balance is achieved.

2) EQUAL BREATHING: It helps to stabilize the emotions and energy body leading to a very relaxed frame of mind. This state of mind is very receptive to either additional information input or effective recall. This is of special help to combat performance stress.

Method: Close your eyes, inhale through the nose on a fixed count - 1, 2, 3, 4. Exhale through the nose on the same fixed count 4, 3, 2, and 1. Repeat the cycle.
These cycles may be repeated for a maximum of 21 rounds at any given time.

3) RAPID BREATHING: This technique is very effective for quickly energizing and relaxing the human system simultaneously. Use of this breathing technique helps to stimulate you when you are in a lazy or dull mood.

Method: Take a deep breath and then breathe in and out in a very rapid manner, in short bursts. (you may count 1, 2, 1, 2…mentally – first at a slow pace and then increase the pace with practice)

4) ALTERNATE BREATHING: Alternate breathing helps to balance the energy flow between two vital energy meridians of the system, restoring the body to a state of balance. This state facilitates the awakening of spiritual consciousness lying deep within.

Method: Place your index and middle finger between the eyebrows. Place your thumb on the right nostril and the ring finger on the left nostril. Gently close your left nostril and inhale through the right nostril. Next, close your right nostril and opening your left nostril exhale through it. Further, inhale through the left nostril. Then, close the left nostril and opening the right nostril exhale through it. Repeat this for seven counts. Start the round by inhaling through the right nostril and terminate the round by exhaling through the right nostril.

5) ABDOMINAL BREATHING: This breathing pattern relaxes and revitalizes you.

Method: Put your hand on the upper abdomen area. Now breathe only through the abdomen while limiting the chest.(you can practice this technique lying down, with one leg folded at the knee and then in sitting posture)

Note: Avoid practising all these breathing techniques two hours after meals.
Avoid Rapid Breathing in case of High B.P and epistaxis.

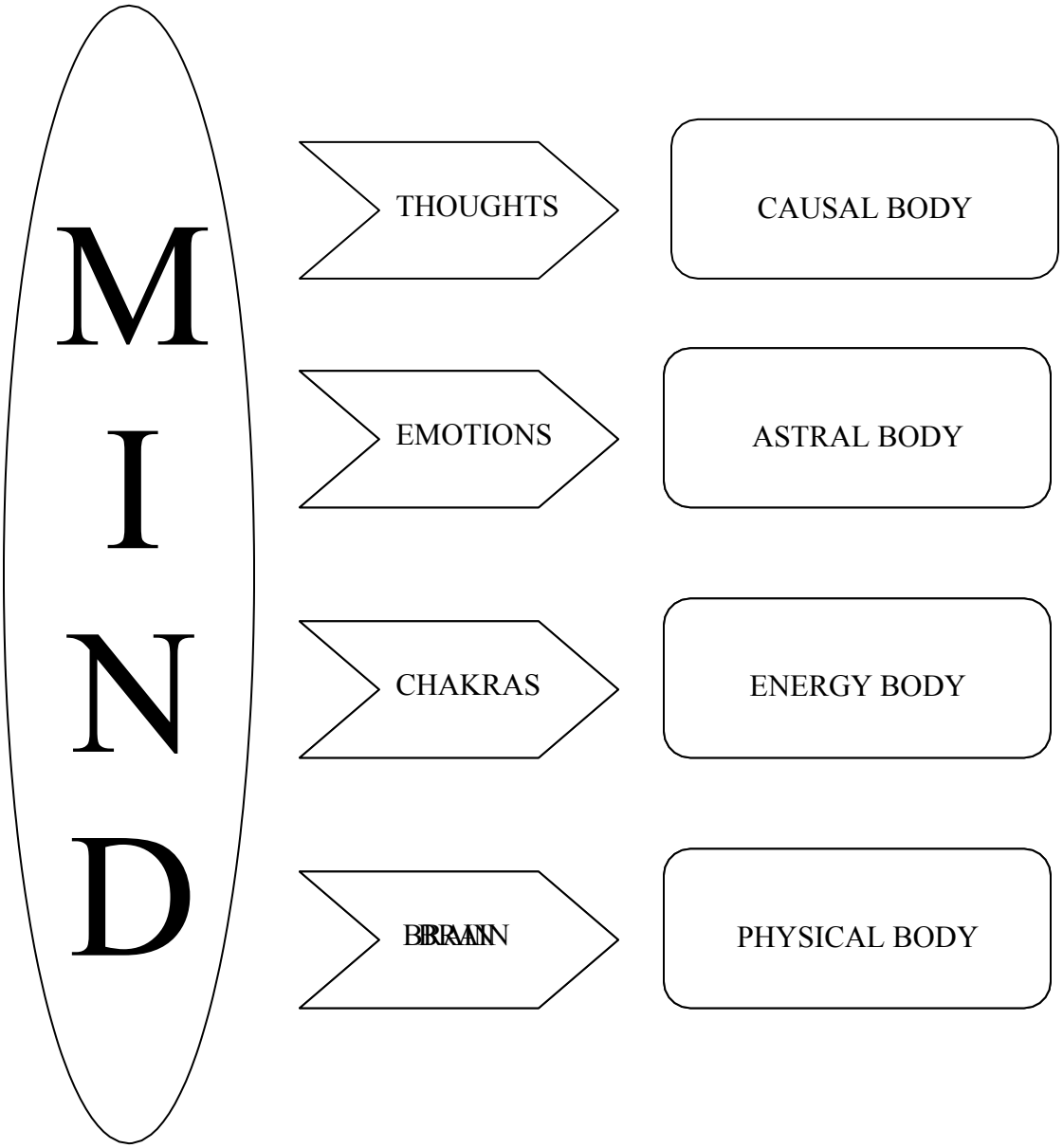
STRETCH EXERCISES

SIMPLE STRETCH EXERCISES TO STIMULATE BODY MERIDIANS AND THE CHAKRAS

Please Note: All the stretch exercises must be done standing in an erect posture.
Complete 7 rounds of each exercise.

1. Start by jerking your hands repeatedly opening and closing the fist, towards the ground. The palms should face the ground. Do this to throw out all the stress, blocks or negativity affecting you.
2. Next, bend your legs from your knees and kick out all the stress, blocks and negativity affecting you.
3. Stand erect with the soles of your feet flat on the floor. Bring both your hands together and interlock the fingers of both the palms. Slowly turn the interlocked palms inside out and gently raise the palms straight upwards, towards the ceiling, till your hands are fully straight and stretched. Now raise yourself on your toes gently and slowly come back down again.
4. Stand erect with the heels of your feet pointing inwards about 2 feet apart. Point the toes as far outward as possible. Place your hands by your sides, turn both the palms till they face the ground and are parallel to the floor. Gently, bend from the knees and come up again, keeping your back absolutely erect.
5. Bring your feet to normal position (feet parallel, with 1 foot gap between them). Now interlock your palms in front of your body and then gently raise your hands and take them to the back of your neck. Don't let the weight of your hands fall on your neck. Gently bend forward from the level of your hips, without bending your knees. Go as far down as possible and exhale while bending forward. Slowly swing back up and as you do this, inhale.
6. With your fingers interlocked and your palms still placed behind your neck, gently rotate your body from left to right, by twisting from the hips. As you turn to each side, exhale and as you come back to the central position, inhale.
7. Keep a comfortable distance between both your feet. Join the tips of your fingers together so that there is no gap between their tips. Now place this finger formation on each of your respective shoulders. Bring both your elbows in front of your chest and try to minimize the distance between them. Slowly rotate your hands outwards i.e. take them upwards and move them in a circular fashion to come back to the starting point.
8. Stretch your arms out in front of you with the palms close to each other. Now spread your arms apart, take them as far back as possible and bring them towards each other again.
9. Stand in a comfortable, erect posture with your hands by your side. Slowly, take your head right back and then gently let it drop down in front of your chest.
10. Gently place your thumbs on the tip of your shoulders and use the rest of your fingers to keep your head steady. Slowly raise your eyeballs upwards, keeping your head steady in the process. Then, gently roll your eyeballs downwards to face the ground. Repeat the eyeball movement from the extreme right to the extreme left.

MIND AND BODY RELATION DIAGRAM



CHAKRAS

HEALING THOUGHTS FOR HEALTHY CHAKRAS

AURA: Aura is the energy field in and around our body. Our physical body is soaked in this energy like sponge soaked in water.

CHAKRAS: The Aura is not uniformly distributed. There are certain centers where energy is concentrated. Since this energy moves like a wheel in a circular fashion, these vortices are called 'chakras'.

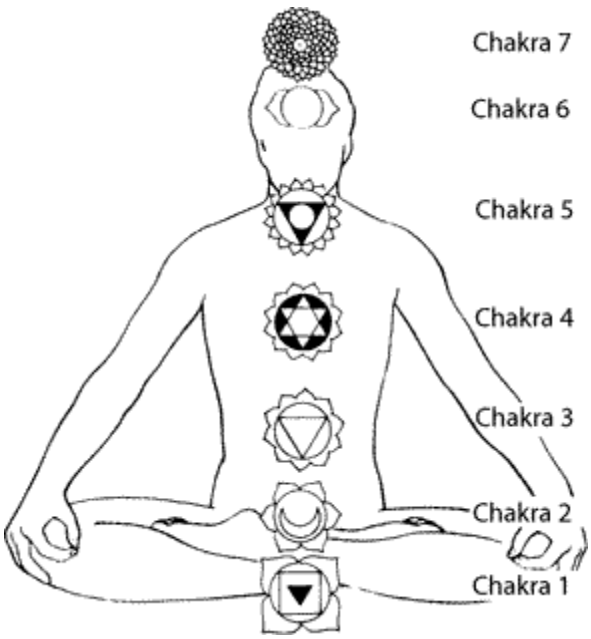
There are 7 major chakras and several minor chakras. The seven major chakras correspond to each endocrine gland.

Functioning of these chakras and the subsequent understanding of the psyche of the person can be detected using crystal pendulums, programmed by the therapists.

ENERGY

The entire universe is a manifestation of energy. This energy is present in all forms that we see on this planet. However, the vibrational frequency of different energy forms is different. E.g. the vibration of energy present in a leaf is very different from the vibration of energy present in a lion, which in turn differs totally from the vibrational pattern of a human being. Similarly, there are various higher vibrations of energy present in the universe whose frequency ranges are much higher than those which can be perceived by the human senses.

Through various processes explained herein represent some ways by which these varied higher energy vibrations can be felt at the human level by the meditator, to enable him to raise his own vibrations, which in turn facilitates a growth in his consciousness.



ROOT CHAKRA

It is a portal through which Earth energy is absorbedinto our system.
It represents that vibration of the Pranik force which aligns our consciousness to the physical living.
This means that the Root Chakra is directly linked to our survival needs on Earth. The most important reaction that we have inherited from the animal kingdom is the 'fight and flight' response. Hence it has a profound effect on loco-motor, circulatory system etc.

Pendulum Movements
Clockwise movement indicates a fully functional root chakra and shows that the person is enjoying his / her physical existence on this planet and putting in his / her best efforts for effective survival.

Anti-clockwise movement indicates a completely blocked root chakra and shows that the person is rejecting the joy of the process of physical existence on this planet. This person could be very depressed and at times exhibits suicidal tendencies.

Pin-point / standstill position indicates a partially blocked root chakra and shows that there are perceived areas of concern about survival. There is a possibility of self- sabotage.

Horizontal movement of the pendulum suggests excessive pressure from higher chakras on the process of survival. It could also be a pressure due to some of the religious beliefs / moral upbringings or values imbibed from society.

Vertical movement of the pendulum suggests tendency of pull between spiritual growth and materialistic needs.

CHAKRA AFFIRMATIONS

I am enjoying every moment of my physical existence on earth. I am open to universal guidance to shape my physical journey. I accept my contract of physical existence on this planet.

SACRAL CHAKRA

This chakra is also known as 'Sex' chakra. It is the most misunderstood chakra because of its name and predominant area of action.
It represents that vibration of the Pranik force which aligns our consciousness to the feeling of joy for all worldly pleasures.
The desire of a mother to touch a child or to hug him is also guided by this chakra.

Pendulum Movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block. This further indicates that there could be a block in the front or back sacral chakra.

- Back sacral chakra block indicates that the person is receptive to worldly pleasures but does not necessarily reciprocate appropriately. These individuals have relationships with many, but find it difficult to commit to any one person.
- Front sacral chakra block indicates that this person could have a compulsive need for multiple sex partners or an overdrive to indulge in worldly pleasures.

Horizontal movement of the pendulum indicates pressure exerted by human need and values imbibed from society.

Vertical movement of the pendulum indicates a pull between the above mentioned aspects.

CHAKRA AFFIRMATIONS

I flow through all my sensuous experiences.
I am open to giving and receiving appropriate sexual energy.
I am open to enjoying all pleasures that the world has to offer to me.

SOLAR PLEXUS CHAKRA

This chakra is a measure of the degree of control you have over daily life.
Fears, anxiety, worry, anger, frustration and all modern day "stress" related issues arise out of this feeling of lack of control.
It represents that vibration of the Pranik force which aligns our consciousness to the feeling of knowing our position in life.

Pendulum movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block. This further indicates that either the front or back solar plexus chakra is blocked.

- If the front is clogged, the person is not receiving energy but emitting from the back which means, such a person willhave a feeling of being helplessly manipulated by the universe. In fact, as a compensatory mechanism the person will try to control many things.
- On the other hand if front chakra is open and back chakra is blocked, the person will be functioning in a structured environment, but their output is disorganized. They find it difficult to take charge of their lives or situations.
- Back solar block comes in the way of outflow of healing energy.

Horizontal movement suggests the feeling of getting pressurized by religious values and spiritual aspects coming down heavily on the perceived need to have adequate control over daily life. There is a feeling of being sandwiched between material needs and the need for higher growth.

Vertical movement suggests a pull between material and spiritual aspects, in day-to-day existence.
Solar is the most important chakra for an attitude ofsurrender.
Ego is a defense mechanism to cover up inner weakness.
The concept of ego sublimation starts at the solar chakra level.

CHAKRA AFFIRMATIONS

I choose to surrender to the flow of universal energy in my life.
All aspects of my life are under appropriate control of the universe. I take charge of my life.

HEART CHAKRA

Heart chakra is the seat of love and acceptance. It is concerned with love for life, living, the Universe, nature, people…all creation. Though a well-functioning solar plexus chakra helps one in becoming a good therapist, a well-functioning heart chakra is necessary for becoming a good healer. All animals operate out of lower three chakras. Human growth starts with the development of the heart chakra.

Pendulum movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block. This further indicates that either the front or back heart chakra is blocked.

- Front block means the person will believe that though he /she is giving love, they never receive enough. They are whiners in life and do not feel loved.

- People with back heart chakra block will keep receiving lot of affection from people but they tend to be insensitive to peoples' needs.

Horizontal movement suggests that the person has not understood the true nature of love and is getting pressurized by his/her spiritual and religious concepts.

Vertical movement, though rarely found, indicates conflict and pull between love life and desire for spiritual/religious growth.

CHAKRA AFFIRMATIONS

I flow in love
I am open to give and receive lots of love.
I forgive myself and all others who have
hurt me. I heal myself and others with love

THROAT CHAKRA

It is related to communication, which in turn is related to self-worth and self-esteem. 'Clairaudience' requires proper development of this chakra.

The Throat Chakra represents that vibration of Pranik force which aligns our consciousness to the feeling of self-belief.

Pendulum movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block. This further indicates that either the front or back throat chakra is blocked.

- Blocked front throat chakra indicates poor listener. Hearing is different from listening. A lot of people hear but not listen.

- People who have a blocked back throat chakra may be able to listen correctly but is not able to express adequately.

Horizontal movement indicates that communication is getting affected by perceived pressure from higher chakras.

Vertical movement indicates that communication is getting affected by a pull towards religious / spiritual aspects of life.

CHAKRA AFFIRMATIONS

I believe in myself
I am open to communication
I am listening precisely and expressing adequately. I'm fine, I'm okay.

THIRD EYE CHAKRA

This chakra is related to higher vision and its implementation. 'Clairvoyance' is related to a fully functional third eye chakra.

The Third Eye Chakra represents that vibration of Pranik force which aligns our consciousness to being visionaries and implementers.

Pendulum movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block. This further indicates that either the front or the back third eye chakra is blocked.

- Blocked front third eye chakra indicates inadequate capacity to creatively visualize.
- People who have a blocked back third eye chakra lack the capacity to implement their dreams and visions.

Horizontal movement of the pendulum indicates the pressure from the crown on demands of the material needs.

Vertical movement indicates a pull between material and spiritual aspect in implementation of dreams and visions.

CHAKRA AFFIRMATIONS

I am open to Universal wisdom

I am ready to implement all my visions

CROWN CHAKRA

This chakra is related to the concept that the world is the manifestation of what one is.
The crown chakra represents that vibration of Pranik force which aligns our consciousness to one-ness with the universe.

Pendulum movements:
Clockwise movement indicates fully functional chakra.

Anticlockwise movement indicates a completely blocked chakra.

Pin point or standstill position indicates a partial block.

- Blocked crown chakra indicates possibility of disintegration due to inner children and a possibility of extraneous energies.

Horizontal movement of the pendulum indicates restricted spiritual growth due to perceived pressure.

Vertical movement indicates a pull between material and spiritual aspect of oneness

CHAKRA AFFIRMATIONS

- *I am open to Universal one-ness*
- *I am ready to heal and integrate myself deeply and completely*

MAJOR CHAKRAS AFFIRMATIONS

POSITION	KEYWORDS	ASSOCIATED FEELINGS/THOUGHTS	AFFIRMATIONS
Between great toes	Aura alignment	Energy body misaligned with physical body. Lack of interest to be in one's body	*I align and integrate my physical and energy body.*
Mooladhar: Base of the spine	Survival	Complete/partial disinterest in material world survival	*I enjoy and completely enroll in the process of life.*
Swadhistana: Near the genitals	Earthly pleasure	Earthly, sensuous and sexual pleasures	*I flow through all my sensuous experiences.*
Manipura: between the navel and epi-gastrium	Control	Lack of control culminating into anger / fear / ego related issues / modern day stress	*I flow with the flow of life.*
Anahata: Center of the chest	Love and Acceptance	Emotional hurt, feeling of being unloved	*I open myself to give and receive unconditional love.*
Vishudhi: Throat	Communication Self Esteem	'Not okay', not being able to express and understand	*I believe in myself.*
Agya: Between the eyebrows	Insight	Inability to conceive or implement ideas / visions	*I implement all my visions; I open myself up to new ideas.*
Crown: Top of the head	One-ness	Duality	*I'm one with the Universe.*

ZONE OF ABUNDANCE AND KARMIC RESPONSE

Our zone of abundance

7 probable zones

- Spiritual
- Mental
- Vocational
- Financial
- Familial
- Social
- Physical

Vibration	Desire	Block	Karmic response
Fight/flight	Fear based	By fear	Tit-for-tat
Power/ego	Ego/senses gratification	Guilt	Hurt ego
Awareness	To take charge of life	Shame	Why me?
Love	Understand higher purpose of life – connect to HS	By grief	This is a karmic debt – I forgive and move on
Sound	Remain connected to the HS or true self	Lies	This is a divine plan I am playing my part
Light	To unite with the divine	Illusion	I have written this part
Oneness	United with the divine	Attachment	Aham brahmasmi

MEDITATIONS

HIGHER SELF MEDITATION

OBJECTIVE: This meditation is aimed at harnessing high vibrational energy from one's own "Higher Self". This energy can be downloaded into the human system through the technique explained and can then be channelized or directed at fulfillment of any professional or work oriented goal.

TECHNIQUE: Sit cross-legged in a comfortable position. Roll both the palms into a fist with the thumb outside the fist. Let the wrists rest comfortably on the knees. The back should be upright. Slowly close your eyes and commence regular breathing. Focus on your breath. Once the system is comfortably settled, tilt your head back slightly so that the forehead appears to be looking at the sky at an angle. Now, visualize a bright, shining star (your Higher Self) about 5 to 6 feet away from your forehead. Feel the powerful, light energy of the star. Take a deep breath. As you are inhaling, simultaneously tighten your fist. As the fist starts tightening, feel the powerful energy of the star entering your forehead area through the third eye (ajna chakra). Once the inhalation is complete, hold the breath for a second or so and then slowly release it. However, while releasing the breath, DO NOT OPEN THE FIST. The pressure on the closed fist must remain. Repeat the process seven times. After that is complete, mentally focus on the goal you wish to achieve. Having visualized it, transfer the energy harnessed at the ajna chakra level to that scene. Then, slowly spread the energy throughout your body and then into the energy field surrounding your body. Slowly open your fists and place your open palms on your heart chakra to complete the meditation.

EARTH MEDITATION

OBJECTIVE: This meditation is aimed at connecting to one's residual energy deposited in the core of the earth. The energy can be absorbed by the human system and can be deployed to different parts of the human body as it has very good healing properties.

TECHNIQUE: Adopt the posture recommended above. Once the system is fully relaxed with regular and rhythmic breathing, take a sharp, rapid breath. As you inhale, tighten the fists. You will feel energy getting absorbed in your system from the base of your spine, which is in contact with the Earth's surface. Hold your breath for a second and then release it slowly. DO NOT OPEN THE FIST. Repeat the process seven times. The energy accumulates at the ajna chakra level. Then slowly direct the energy to flow to the area of your body which requires healing. Slowly open your fists and carry your open palms to your heart chakra to complete the meditation.

FORGIVENESS MEDITATION

OBJECTIVE: The meditation helps in dissolving pains, hatred, and differences in inter-personal relationships thereby enabling an individual to reach a state of equilibrium in his emotional profile.

TECHNIQUE: Adopt the posture described above. Sit still in chin mudra state. Slowly, as you inhale and exhale rhythmically, visualize the person you are closest to and feel most comfortable with. Feel all the love and tenderness you have for this person. Let the energy and emotion fill your heart chakra.

Then slowly replace the image of this person in your mind with the image of a person you are most uncomfortable with. Imagine this person in a big, black bubble approaching you. Use all the love energy accumulated by you at the heart chakra level and direct it to this bubble. Feel the bubble exploding and the person you are uncomfortable with approaching you. Hug the person in your mind. Forgive him and seek his forgiveness in return. Slowly open your palms and place them on your heart chakra to complete the meditation.

HEALING MEDITATION

OBJECTIVE: To heal and shield affected body parts and energize the energy skeleton.

TECHNIQUE: This meditation can be done in a lying down or sitting position.
Take three deep breaths and say the words "RELAX" silently to yourself as you exhale.
Imagine or visualize that you are standing in front of two bath tubs, each one of a size 7'x3'x'3'.
One tub is full of crystal clear healing liquid. It is pure and gentle like spring water collected from a snow-clad mountain.

A]

- Separate the affected part from the body and dip it in this liquid. If all parts appear to be fine, the exercise can be done with any organ such as the heart, liver, or kidneys.
- The moment you dip it in the liquid, see various colors coming out and dissolving in it. All these colours represent various toxins and accumulated negative thought forms. Some of you may have difficulties visualizing these colors. In this case go ahead and imagine or simply feel the difference in the physical sensations of the affected part dipped in the healing liquid. Allow this part to remain in the healing liquid for three minutes or till the time that it becomes crystal clear again.
- Take this part out of the bathtub…
- Now, visualize or imagine golden light falling from the higher dimensions. Hold this part in your hands and bask it in the golden light. Golden light is energizing each and every cell of the body part, drying it in the process.
- Now dip this part into the liquid gold filled in the adjacent tub. This liquid gold forms a protective film all around the part. This part is now cleansed, energized and shielded, ready for replacement back into the body. Replace the part in its appropriate place.

B]

- Now visualize or imagine your energy skeleton separating from the main body. This energy skeleton consists of all the chakras and the central energy line starting from the base of the spine, ending into the energy center in-between eyebrows, supporting the various chakras and two more energy lines intertwining across the central energy line and ascending upwards to end in each of the nostrils (Ida and Pingla). Put this energy skeleton now into the healing liquid.
- Once again, see various colors coming out from various charkas and various energy lines. In case you are not able to visualize, just wait for approximately three minutes or as much time as your intuitive mind guides. Once this process is over, take the chakra skeleton out and bask it in the golden light for energization. See the chakras spinning again
- (Energy skeleton is the only body component we would avoid putting into the liquid gold in the second bathtub.)
- Now replace this clean and energized energy skeleton into the body once again.

C.

- Now visualize the entire body dipped into the healing liquid. See various colors emanating from it as you release all your pain, stress and trauma…feel the relaxation.
- Remain in the tub till the liquid gets clear.
- Now, step out from the tub into the golden light falling from the universe. Dry yourself and feel the entire body being energized.
- Finally, immerse your entire body in the adjacent tub of liquid gold for a protective shield to be coated on it. Stay there for a while..
- Feel your entire system cleaned, energized, shielded and full of vigor and vitality.
- Watch your breath, which has become deep and relaxed.
- Thank master Alton and all the other Ascended Masters for guiding you through this healing. Thank your Angel Guides, and last but not the least Dear Mother Earth and Universal Energy for guiding you through this process of healing and love.

VIOLET BUBBLE MEDITATION

OBJECTIVE: To release all trapped and deep-rooted negative emotions and thought forms.

TECHNIQUE: This meditation is preferably done sitting on any flat surface with crossed legs or on a chair with feet firmly grounded on the floor. Close your eyes.

- Take three deep breaths and slowly tell yourself "RELAX!" silently, as you exhale.
- Imagine a shaft of white light coming from the Universe above and entering the crown chakra, going down through your spine, and exiting from your tail bone, penetrating deep down into the centre of Mother Earth.
- Give an intention to remove all the negativity that you have gathered throughout the day and in the past.
- Now, create a violet coloured balloon around you surrounding your entire body.
- Gently breathe into this balloon…slow, gentle rhythmic breathing…in & out…in & out and with each breath feel your entire body relaxing.
- Now, tell your mind to go to the POINT OF ENTRY of the current problem in your life.
- You may either get some visions, experience some body pain or you may feel some emotional discomfort…Be with it…Just become an observer.
- Now, silently give this intention three times "**I INTEND TO RELEASE ALL MY PAST PAINFUL AND NEGATIVE CELLULAR MEMORIES TO THE WHITE LIGHT OF THE HOLY SPIRIT RIGHT NOW**".
- Breathe rapidly into the violet colored balloon…1-2-1-2-1-2-1-2...Give it your best and with every breath you are releasing all your accumulated negativity into this balloon
- When you feel tired, gently slow down your breath.
- The air in the balloon is thick and cloudy with all your released negativity.
- Now, visualize white light falling, on the balloon from the universe above. See the balloon bursting completely into tiny particles, releasing all the accumulated negativity to the divine forces.
- Feel your body becoming lighter; a kind of vacuum is created. Now think of a happy event in your life. Anything you feel very happy about…a thought, a relationship, or an object.
- Gather all these feelings and happy memories in the form of pink light in the heart chakra. From the heart chakra spread this feeling of love and joy to the various parts of the body.
- Be with your happier self.
- Thank your Angel Guides, Healing Angels, Masters, Dear Mother Earth, and Universal Energy for guiding you through this mediation.
- Whenever you want you can open your eyes with a big smile.

(It is advisable to gently massage all the body parts with a lot of love, to spread this energy uniformly in all the parts of the body.)

EMOTIONMANAGEMENTTECHNIQUE (EMT)

When Is EMT Useful?

- Disturbing emotions like fear, anger, worry, tension, anxiety, depression, hatred, jealousy, insult, frustration etc.
- Physical aches & pains.
- Understanding the core emotion.

PRINCIPLE:

Thoughts, emotions, energy levels are interlinked & known to influence the body & vice versa.

There is close association of emotions & energy. Enthusiasm brings in the energy & zing in the energy system, on the contrary despair saps the energy out of the system. Thus with any emotional disturbance is accompanying distortion of the corresponding energy lines. Once the emotional stability is attained, the energy lines are restored to their normal state & vice versa.

By the Concept of Reversibility & Law of 'Similars':

We can reverse the energy disturbance brought on by disturbed emotional state by creating a fresh external artificial similar disturbance at the level of the energy. This is done by gently stimulating certain energy points. In an attempt to overcome the artificially created disturbance, the person's vitality sets the system right & the corresponding emotions are balanced.

TECHNIQUE:

1. Pay attention to the problem in your body or mind (Example: Pain or anger etc).
2. Get in touch with the emotions/feelings associated with the complaint/problem. (Example: 'Headache after feeling upset following an argument' OR 'Angry when things don't go your way')
3. Try & be specific in assessing the feeling. The clearer the feeling becomes, quicker one finds the relief. Nevertheless the technique works but may take longer.
4. Now give this emotion or complaint a number on scale of 0 - 10, depending on the intensity & severity, zero being no pain or complaint & 10 being the maximum in severity.
   (You may write down the feeling/complaint with the grade in front of it, for your reference as you work on it further.)
5. Making the Affirmation:
   "Even though I feel/have_____, I choose to heal & integrate that part of me".

Examples:

(a) Even though I have *this fear of exams*, still I choose to heal & integrate that part of me
(b) Even though I have *this fear of failure*, still I choose to heal & integrate that part of me.
(c) Even though I have *these nightmares*, still I choose to heal & integrate that part of me.
(d) Even though I have *this headache*, still I choose to heal & integrate that part of

6.

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me.

(a) Even though I have *this stiffness in my neck*, still I choose to heal & integrate that part of me.
(b) Even though I have *this craving for junk food/chocolates*, still I choose to heal & integrate that part of me.
(c) Even though I *am feeling very upset as my teacher scolded me*, still I choose to heal & integrate that part of me.
(d) Even though I *am feeling angry towards my mother*, still I choose to heal & integrate that part of me.
(e) Even though I *don't like XYZ subject*, still I choose to heal & integrate that part of me.

7. Repeat the affirmation ***Three Times***, ***loudly with emphasis***, while gently rubbing / tapping the edge of the palm (Karate chop area) OR knee joints.
8. Then, gently rub the *thymus* point (in the middle of the breast bone) for the count of seven breaths.
9. CLOSURE: Close your eyes and think of something or someone that you love.
10. EFFECTS:

1) If the problem is ***relieved totally*** i.e. come down to '0' on the scale of 0-10, then go on to the next emotion / feeling that may crop up.

2) If the problem is***relievedpartially***, then mentally give it a number between 0 - 10, & start another round of EM saying the **slightly modified** affirmation i.e.

*"Even though there **is remaining**_____, I choose to heal and integrate this part of me".*

While rubbing/tapping the end points of energy meridian, say "remaining_____"

(Example: "Even though I have this remaining headache, I choose to heal & integrate this part of me" OR "Even though I have this remaining anger, I choose to heal & integrate this part of me")

3) If there is ***no change at all***, your subconscious mind is probably **resisting** the change. In such a situation do the EMT by saying *"Even though **I don't want to get over**_____, I choose to heal & integrate this part of me.*

(Example: "Even though I don't want to get over my headache, I choose to heal & integrate this part of me" OR "Even though I don't want to get over my anger, I choose to heal & integrate this part of me"")

NOTE:

- *You can use any side of the body for stimulating the end points of meridians.*
- *Say the affirmation loudly with feeling & emphasis rather than saying it mechanically.*
- *EMT can be done anywhere, anytime & any number of times till there are no issues or problems.*
- **Yawning, sighing, coughing, passing gas are very commonly seen & indicate that this technique is working effectively.**
- **This technique is based on the research work done by GARY CRAIG from U.S.A. and has been modified based on client feedback.**
- **ENJOY THE FREEDOM FROM RESTRICTING EMOTIONS & EMPOWER YOURSELF!!!**

GUIDELINES FOR SMOKING CESSATION

1. Case receiving – get details of the habit
2. Teach EMT for craving and withdrawal symptoms. If the client has attempted giving up the habit before this, they are very well aware of the withdrawal symptoms. Neutralize the emotions.
3. Ego States – for attachments/IC work/PAB/metaphorical representations
4. Re-inforcement with positive suggestions (Self hypnosis)
5. Script for will power x 21 days

COUNSELING & INTERVIEWING

COMMON PITFALLS OF INTERVIEWING

Three bad styles of interviewing:

- Over sympathetic or empathetic
- Asking too many questions ("Just the facts")
- Need to make deep psychological interpretation right away.

A new therapist has an inherent idea of what his/her role is and will then cause one of the three problems listed above.

CLINICAL INTERVIEW

The Clinical Interview is also called the "Psychiatric Interview", "Intake", and other names. Here it refers to the initial contact(s) with the client that is aimed at gathering information so that a diagnosis and appropriate Treatment Plan can be formulated. Except for the "Presenting Problem", most of the Clinical Interview is focused on historical information about the client.

IDENTIFYING DATA

Identifying data include name, age, date of birth (actual, not legal), marital status, education level, ethnic group, occupation, home address, phone numbers, email, etc.

PRESENTING PROBLEM

Use the client's own words to the degree possible. Make sure the following is included:
Symptoms reported by client
When the problem began
Course of the problem
Precipitating event
Coping strategies of the client
Ever had this problem before?
Behavioral description of problem (Especially for child clients).

BACKGROUND INFORMATION (OR CLIENT HISTORY)

Family History:

- Ask about original family
- Who and how many? Father, mother, brothers, sisters?
- Family atmosphere – happy, tense, punitive, etc.
- How did you get along with family members?
- One closest to and why
- One farthest from and why
- Ask about current family
- Who and how many? Married, children, others living at home.
- Family atmosphere – happy, tense, etc.
- Dynamics – who is close to whom?

Developmental history of client:

- Pre-natal and birth information (if known to client)
- Pre-school years
- Developmental milestone information (late or early)
- What do you think those years were like for you?
- Any unusual trauma – physical, psychological, familial?

Elementary school years (grades K-6)

- Grades client made
- Teachers especially liked or disliked
- Any unusual stresses or problem periods: at home, school, with friends?

Adolescence (Junior high school)

- Academic factors
- Friendship patterns – How many friends? Popular? Unpopular?
- Relationships with opposite sex
- Attitudes toward sex
- Any "rough times" psychologically?
- Any problems? Treatments received?
- Any unusual stresses – school, home, family, medical, etc.

High school years

- Cover the same factors as above for Junior High
- Individuation: Did you begin to break away from family?
- What effect did this have on family?

College years (if applicable)

- Academic factors
- Characterize these years
- Any unusual stresses – school, home, psychological, etc.?

Marriage or other life style

- What did you like about mate when dating?
- How do you feel about marriage (or lifestyle) currently?

Occupational history

- What job or jobs have you held?
- How do you feel about your current job?
- Any unusual stresses at work?

Financial history

- Any unusual stresses in this area?

SPECIFIC AREAS TO EXPLORE IN DEPTH

Medical and Psychiatric History: Get thorough history here – when, where, precipitating events, drugs etc.

Any other area needing in-depth exploration: Example, if client is seeking help for a marital problem, therapist would need to go into depth, taking history about dating, original reasons for marriage, course of events to date, and primary stresses in relationship.

SUMMARIZING & ENDING THE INTERVIEW

Ask: "Is there anything else at all that has occurred or is occurring in your life that might better help me to understand the problem?"

Goals of the client: "What would you like to work on, and what would be the goal?"

Summarize and formulate the problem, and make recommendations for treatment, further testing etc. Explain the next step carefully to the client.

PRE-REQUISITES OF A GOOD THERAPIST

- Intention to heal
- Intention to be a pure channel
- Non-biased and non-judgmental approach to the client
- Privacy
- Patience, Perseverance & Tolerance
- Energy exchange
- Detachment to the result
- No criticism of other therapeutic practices
- Disclaimer

WHY DO WE HEAL

- Monetary considerations
- Positive Ego satisfaction
- Desire to operate through the higher self, altruistic motive
- Experience of being a channel of Universal Energy
- Clearing Karma

REASONS FOR OCCASIONAL FAILURE

- First encounter was not good-refer to another therapist.
- Self doubt about healing capability.
- Client prefers remaining sick i.e. psychological resistance.
- Root cause of problem not tackled.
- Karmic issues-Mind says time is not ripe
- Past life cellular memories

THERAPIST QUESTIONNAIRE FOR SELF WORK

1) My motive for completing my courses has been met? Yes / No
   If no, please write the reasons:

2) Rate the following on a scale of 1 to 5.
   1 = never, 2 = rarely, 3 = sometimes, 4 = mostly, 5 = absolutely / strongly

I love conducting therapeutic sessions for others	
I have complete faith in the healing process	
I have to make money from the sessions	
I feel my client is appraising my skills	
I have to follow up with the clients to ask how they are faring	
I have a fixed plan of action for therapy	
I also need therapy	
I compare myself to other therapists	
I am sure I have complete understanding of the techniques I have learnt	

3) My intention for every session that I conduct is:

4) I feel that the following could be a window to some of the belief systems I recognize with regard to these therapies / spirituality:
   - My family always maintained / said:
   - My mother made me feel / realize:
   - My father was of the opinion:
   - My spouse feels / says:
   - My friends feel I am:

5) According to me, an ideal therapist is / should be:

ANATOMYANDPHYSIOLOGY

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COMPLETE INDEX FOR ANATOMY & PHYSIOLOGY

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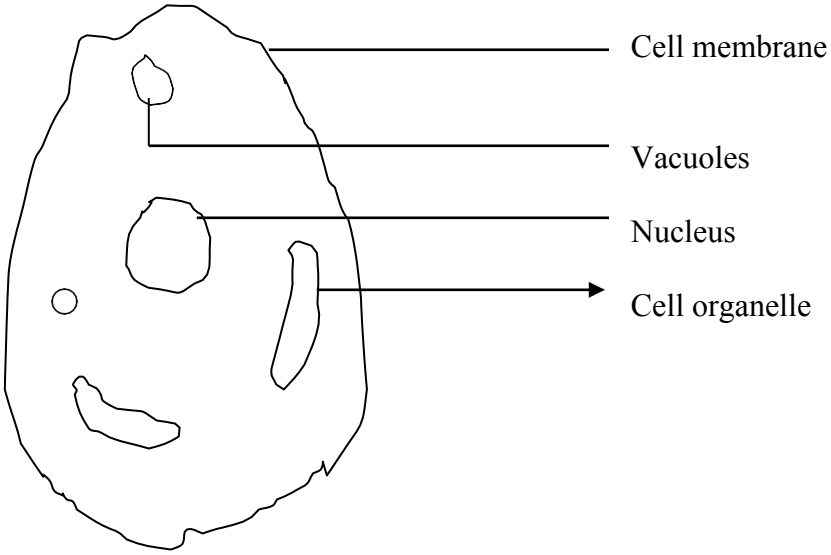
Chapter 1
General Pathology

General Terminology including general symptoms:

SUFFIX: -itis	Inflammation e.g. Gastritis, Sinusitis, Bronchitis, Gingivitis
-otomy	To open and close a hollow organ e.g. Laprotomy(oprning up the abdomen)
-ectomy	Removal from body e.g. Hysterectomy, Appendectomy
-oma	Tumour e.g. Lipoma, Carcinoma
-lysis	Destruction e.g Autolysis (auto destruction of the cell) , Spondylolysis
-algea	Pain e.g. Arthalgia (joint pain), myelgia (muscle pain)
-pathy	Pathology e.g. Cardiomyopathy, Nephropathy
-rrohea	Discharge e.g. Rhinorroea, Diarrohea
-scopy	To see
PREFIX: Leuco	White
Nephro	Kidney
Hepato	Liver
Hemato	Blood
Neuro	Nerve and brain
Gastro	Digestive system
Arthro	Joints
Osteo	Bones
Myo	Muscles
Chondro	Cartilage
Endo	Hormonal
Pulmo	lungs
- A	No
-Multi	Many
Extra	Outer
Intra	Inner
Inter	Between
Auto	Self
Hyper	More

Hypo	Less
Heamo	Blood
Dermato	Skin
Diagnosis	Identification of disease
Prognosis	The probable course of the disease
Oedima	Swelling
Haemorrage	Bleeding
Ulcer	A localized breach(gap) in the surface continuity –skin or mucosa
Atrophy	Diminution in the size
Dystrophy	Diminution in the size due to lack of nutrition
Hypertrophy	Increase in the size without any increase in no.
Hyperplasia	Increase in size due to increase in no. of cells
Hypoplasia	Incomplete development
Aplasia	Failure of development
Paralysis	Loss of motor(movement) power
Hemiplagia	Paralysis of one half of the body
Paraplegia	Paralysis of both the lowe limb
Quadriplegia	Paralysis of all four limbs
Metastasis	Spread of a local disease (like the cancer) to the distant part of the body
Epithelium	Outer cell lining – The outer most layer

CELL STRUCTURE



CELL

Cell :

- Cells are the microscopic building block of body.
- Cell contains cytoplasm, nucleus, and various cell organelles
- Cell is covered by a semi-permeable membrane called 'cell membrane'
- Inward or outward flow of fluid and matter are controlled by cell membrane.
- There are various 'junctions' , 'pumps' and 'channels' which control the electrolyte and fluid flow inside and outside the cell
- Body is broadly divided into 'extra cellular' and 'intracellular' compartments. The content of these compartments are different

Blood:

Vennules

Blood has two components

- Serum- made up of protein and lymphatic fluid mainly
- Corpuscle- RBC , WBC and PLATELETS

Function of each components:

- Serum- the matrix of blood , maintain fluid dynamics
- Corpuscle-
  - RBC : carry oxygen
  - WBC : (leukocyte) Help in body defense
  - Platelets: helps in healing

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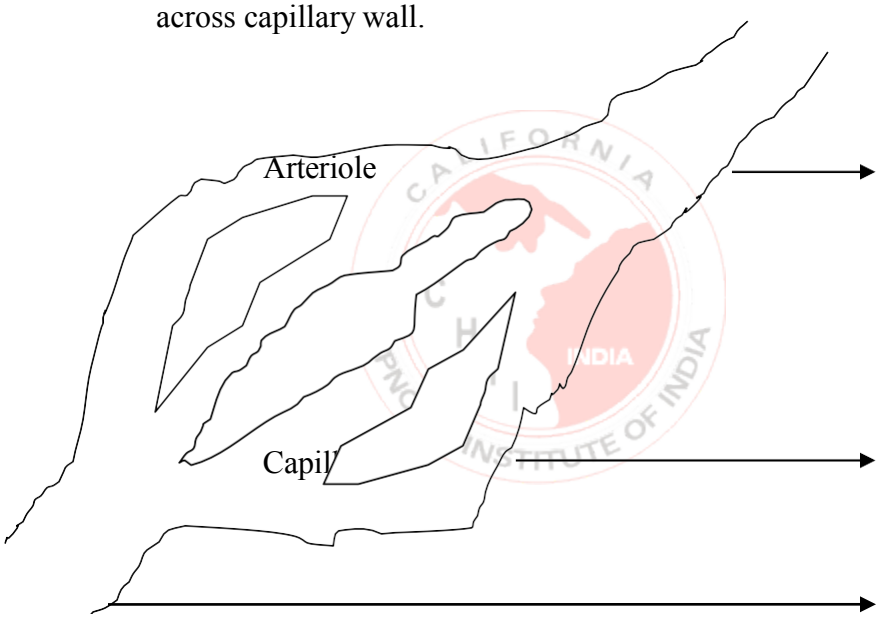
Blood Supply:

- Artery : Carries blood from the heart .Carries oxygenated blood ,exception: pulmonary artery
- Vein: Carries blood to heart. Carries deoxygenated blood exception: pulmonary artery.
- Capillary: Junction between artery and vein- Exchange of metabolites and nutrients happen across capillary wall.

Lymphatic System : Lymphatic system comprises



1. lymph vessels
2. central lymphoid tissue
3. peripheral lymphatic organ
4. circulatory lymphocytes

Functions of lymphatic system:

- Primarily responsible for coarse drainage
  -Remove large protein molecules and other particles from the tissue space
- Production and maturation of lymphocytes

Cell Injury

- Reversible (cell swelling & fatty changes)
- Irreversible

The cellular response to injurious stimuli depends upon types of injury, duration and its severity.

Causes:

1. O2 deprivation (hypoxia)
2. Physical agents
3. Chemical agents and drugs
4. Infectious agents
5. Immunologic Reactions
6. Genetic Derangements
7. Nutritional Imbalance

Ischemic Cell Injury: Cell injury due to decreased blood supply

Infarction: Cell Death resulting from occlusion of blood supply

Necrosis: Series of events leading to cell death

Gangrene: Putrefaction of necrosed tissue

Inflammation:
First response of the body to trauma or foreign body
There are five signs of inflammation

Redness
Swelling
Pain
Rise in temperature of that part
Loss of Function
Acute- is an immediate response to injury or exposure to offending agent.
Neutrophils are the predominant cells.

Increased Blood Flow → Arteriolar Dilatation (redness) → Opening of Capillary bed →Increases Vascular permeability → Accumulation of fluid and WBCs in the extra vascular space → Phagocytosis

Chronic- is an inflammation of prolonged duration in which active inflammation, tissue destruction and attempts at repair are proceeding simultaneously.

Chronic Inflammation arises under the following settings

1. Persistent Infection
2. Prolonged exposure to potentially toxic agents
3. Autoimmunity

Immunity

Innate-
that you are born with. It comprises of Specific types of WBCs e.g. Neutrophils, Monocytes, and Eosinophils.
Acquired-
that which you acquire e.g. through vaccination. It includes Lymphocytes, and complement system.

Wound Healing and Tissue Repair

Factors affecting Healing

1. Nutrition
2. Metabolic status
3. Circulatory Status
4. Hormones
5. Infections
6. Mechanical Factors
7. Foreign Bodies
8. Size, location and type of the wound

Fibrosis (repair by connective tissue)
During irreversible cell injury, the affected part is replaced by fibrous tissue

Neoplasia

Any new growth is called Neoplasia.
It may be benign or malignant

Stages and grades of various malignancies:

BENIGN	MALIGNANT
1. Well differentiated	1. Loss of cell structure
2. Grows slowly	2. Grows rapidly
3. Does not invade surrounding tissue — therefore mobile	3. Invades surrounding tissues —therefore fixed
4. Does not metastasize(spread)	4. Metastasizes

There are various ways to grade a malignancy depending upon the invasion and metastasis, e.g. TNM classifications (considering three parameters T- tumors, N- nodes, M- metastasis)

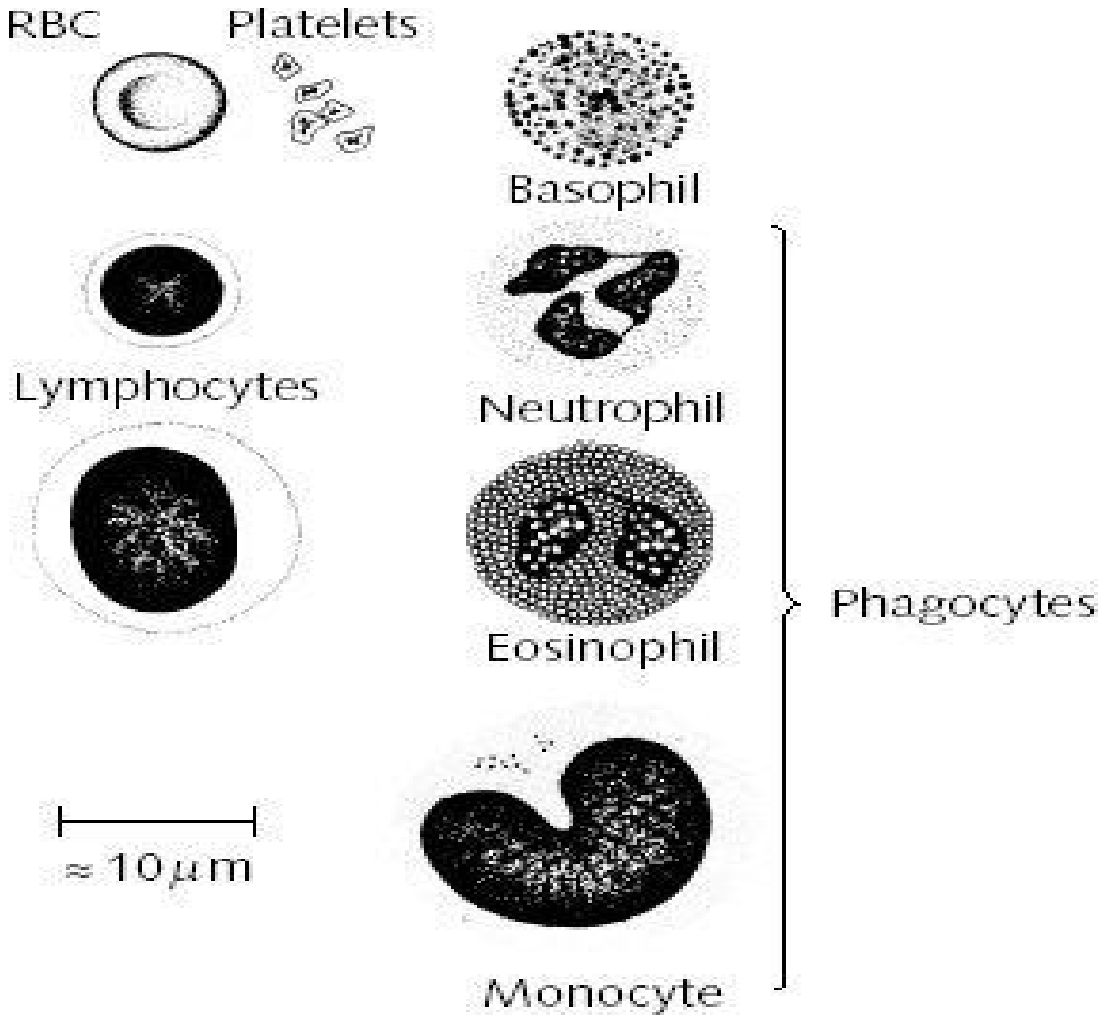
Metastasis- Distal spread of malignancy. Also called as "secondaries"
Dysplasia - Abnormal alteration in shape and size of cells – loss in uniformity of cells
Metaplasia- Loss of differentiation affecting a tissue

Chapter 2

Cardiovascular System

BLOOD:

- RBC carries oxygen
- WBC helps in inflammatory response and other immunity response
- Platelets and 8 other 'intrinsic factors'(secreted from platelets or free floating in blood serum) are responsible for stoppage of bleeding from an injury
- All blood cells are derived from haemopoetic cells in the bone marrow
- Erythropoietin is the hormone produced in kidney ,responsible for RBC production
- Lymphocytes are derived from lymphoid tissue in bone marrow
- Vit B12 and folic acids are important for maturation of RBC
- Neutrophil, macrophages, Lymphocytes and eosinophils are main WBC taking part in immunity.



Anaemias:

Deficiency of red blood cells.

IRON DEFICIENCY ANAEMIA:
As iron is necessary for RBC maturation and Haemoglobin synthesis, deficiency of iron causes anaemia.

BLOOD LOSS:
Rapid blood loss due to trauma or other causes (hookworm, piles, heavy menstruation) can cause anaemia

APLASTIC ANAEMIA:
Aplastic anaemia is the result of non functioning bonemarrow

MEGALOBLASTIC ANAEMIA:
Anaemia because of Vit B12 and folic acid deficiency. The RBC look bigger and densely coloured (Macrocytic hyperchromic)

HAEMOLYTIC ANAEMIAS:
Anaemia because of defect in RBC.

1) Thallesemia:

Defect in " GLOBIN" (the polypeptide chain of haemoglobin) chain. RBC becomes fragile and short lived.

2) Sickle Cell Anaemia:

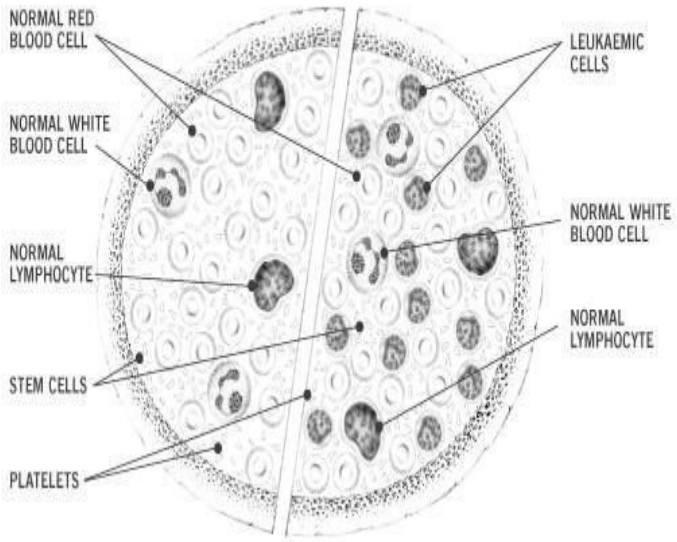
Defect in cell morphology. The oxygen carrying capacity is greatly reduced and RBC becomes extremely fragile.

ROLE OF WBC IN INFLAMMATORY RESPONSE:

- The tissue macrophage is the first line of defense against invading organism (PHAGOCYTOSIS)
- Neutrophil invasion is the 2nd line of defense
- Invasion of macrophage is the 3rd line of defense
- Mass production of WBC is the 4th line
- Mass production of eosinophils in case of parasitic infection as they are large they can't be phagocytosed . Eosinophils get attached to them and secrets lytic substances

LEUKEMIAS: (Blood Cancer)

- Uncontrollable cancerous production of WBC.
- Either Bone marrow or other lymphatic tissue is responsible
- The rapidly growing cells invade tissues and use the metabolic elements of these tissue causing tissue destruction by metabolic starvation (nutrient depletion)



IMMUNITY

- **Innate immunity:**
  - Generalised action like phagocytosis, acid production, skin defense etc. Its present from birth. 1st line of defence

- **Acquired immunity:**
  - Ability to develop extremely complex and powerful and complex defence mechanism to a specific agent
  - There is formation of 'memory' cell which stores the memory of a specific antigen(toxin), so at the 2nd exposure there are preformed specific defence mechanism towards that toxin

ANTIGEN-ANTIBODY:

Antigen :
It's the foreign body (normally protein) which gives rise to antibody production.

Antibody:
Secreted by plasma cell these are the specific protein meant to destroy the antigen.

ANTIGEN ANTIBODY REACTION:

When antigen and the specific antibody for that antigen comes together they gives rise to antigen antibody reaction by which the toxicity of the antigen is negated.
The reactions are:

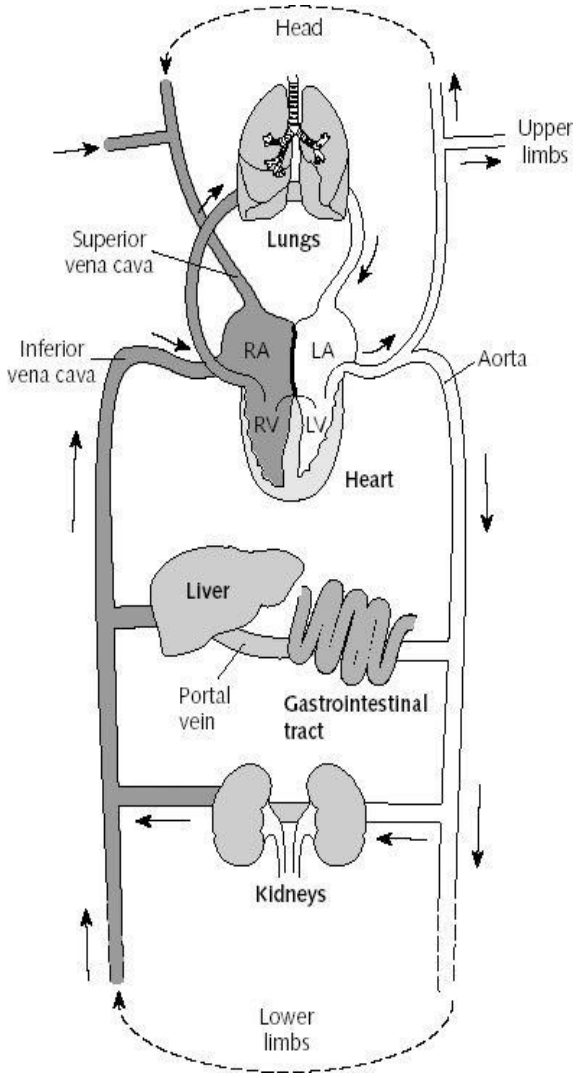
- Agglutination
- Precipitations
- Neutralizations
- Lysis

ALLERGY AND HYPERSENSITIVITY:

Sometimes hyper immune response cause allergy or other types of hypersensitivity reactions.

All allergies, asthma, hey fever, anaphylactic shock are examples of hyper immune response

CARDIOVASCULAR SYSTEM



CVS:

- Heart is the 'pump station' for blood circulation having 4 chambers, 2 atriums and 2 ventricles
- Heart beats at a particular rhythm due to impulses generated at 'SA node'
- Left ventricle supply oxygenated blood (WHITE IN DIAGRAM) to the whole body
- From the body de oxygenated-bad blood (BLACK IN DIAGRAM) comes back to the right atrium
- From the right atrium blood travels to the right ventricle →and goes to the lungs
- Gets the oxygen→ comes back to left atrium→ left ventricle
- The valve between left atrium and left ventricle called 'mitral valve'
- The valve between right atrium and right ventricle called 'tricuspid valve'

DISEASE OF HEART RHYTHM:

- Bradycardia: Slow heart rate, less than 60 /min
- Tachycardia: Rapid heart rate , more than 100/min
- Heart block:
  When electromagnetic impulses are blocked due to ischemia or other factors. There is disturbance in the heart rhythm it may become irregular, or miss a bit at regular interval (partial block) or can completely stop the heart (complete block)

- **Ventricular fibrillation:**
  - Kind of 'epilepsy' of heart due to erratic electromagnetic impulse generate .
  - Rapid random contraction of heart.
  - Heart appears like a bag full of worms

- **Atrial fibrillation: Atrium contracts randomly**

CONGENITAL HEART DISEASE:

- **Congenital heart disease is basically anomaly (defects) of heart anatomy from birth**

- **VSD:**
  - Ventricular Septal Defect
  - Hole in the septum dividing two ventricles . Oxygenated and deoxygenated blood are mixed

- **ASD**
  - Atrial septal defect

- **Valvular defects:**
  - Defects in the valve, stenosis, prolepses etc

RHEUMATIC HEART DISEASE: (RHD)

Rheumatic Heart Disease (RHD) is followed by Rheumatic fever. At first there is streptococcal bacterial infection (SORE THROAT)→ fever→ antibody produced against them→antibody attacks the heart valve tissue and destroys them. Sometimes heart valves are fibrosed or calcified.

MIOCARDIAL INFARCCTION (MI): ISCHEMIC HEART DISEASE (IHD):

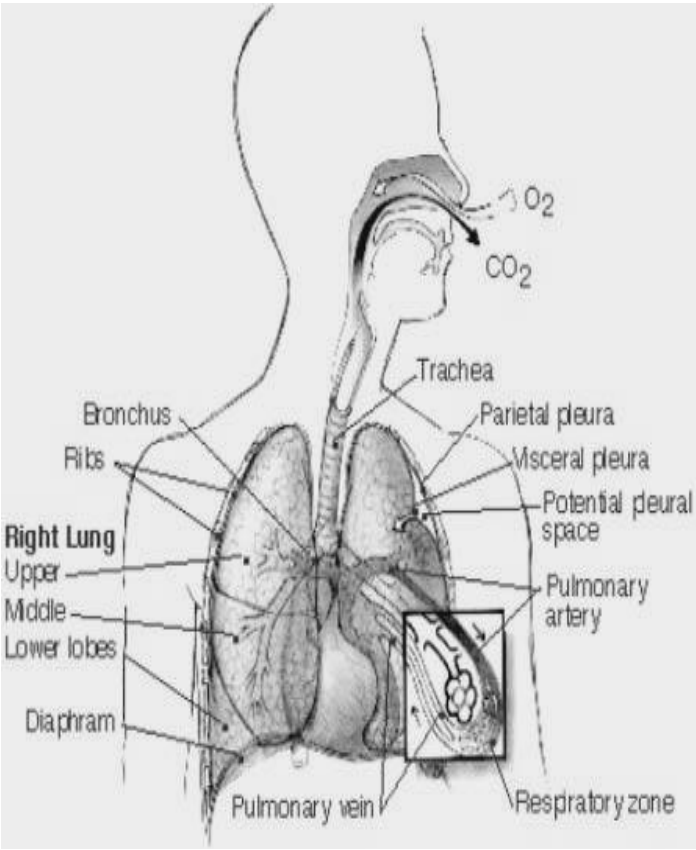
The blood supply to the heart is hampered because of the blockage of arteries that supply the heart muscles. If cells are irreversibly injured its called INFARCTION. If the injury is reversible it is called ISCHEMIA

The pain experienced due to ischemia of heart is called ANGINA

Chapter 3

Respiratory System

Respiratory system:
Respiratory system can be classified according to structure and functions.



- Structurally:

  1. The Upper Respiratory Tract
     - nose
     - pharynx
     - associated structures

  2. The Lower Respiratory Tract
     - larynx
     - trachea
     - bronchi
     - lungs

- Functionally:

1. Conducting Zone

2. Respiratory Zone

SYMPTOMS OF RESPIRATORY SYSTEM:

DYSPNOEA:
Breathlessness or dyspnoea can be defined as

"Unpleasant subjective awareness of breathing" commonly seen in respiratory diseases and cardiac diseases.

The Respiratory causes of dyspnoea:
-Chronic Obstructive Air way Diseases (e.g. asthma)
- Restrictive Lung Diseases (e.g. Pneumonia)

HAEMOPTYSIS:

"Blood coming out with cough"
Causes of haemoptysis:

- Bronchial Disease:

  -Carcinoma
  -Bronchiectasis
  - Acute Bronchitis
  -Foreign body

- Parenchymal Disease:
  -Tuberculosis
  -Pneumonia
  -Lung abscess
  -Trauma
- Lung Vascular Disease:
  -Pulmonary embolus
  -Infarction

Upper Respiratory Tract Infections:

'Common cold' is the most common.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE:

Chronic Bronchitis:
Clinically can be defined as a disease where sputum is being coughed up on most days of at least three consecutive months for more than two successive years.
Causes:

- Irritant (smoking, poisonous gas etc)
- Pollutions, Dust, Smoke, Fumes
- Infections (may be super added)

Hypertrophy of the mucous producing gland causes increased production of mucous.
→Mucosal edema→ Air flow obstruction

Emphysema: Dilatation of alveoli
Causes:

- Smoking
- Other Irritants

Persistent inflammation of the airways and the alveolar septa causes irreparable damage to the surrounding connective tissue

Bronchial Asthma:

The symptoms of bronchial asthma are caused by an inflammatory reactions within the bronchial wall which makes the bronchi narrow.

Caus
es:

- pollen
- mites in house dust
- feather
-

- drugs
- foods
- industrial chemicals

Types:
1. Early onset- since childhood due to allergens
2. late onset- adults (intrinsic asthma): infection, exercise, environment, emotions

Bronchiectasis:
Abnormal dilatation of the bronchi.
Causes:
- Severe infections
- TB
- Foreign Body

RESTRICTIVE LUNG DISEASE:

Pneumonia:
Pneumonia is the term used, to describe inflammation of lung tissue.
Cause :
- Infections (Streptococcus pneumoniae is the commonest)
- Aspirations
- Inhalation of toxic gas

Tuberculosis:
Pulmonary TB causing lung fibrosis and eventually lung collapse.

Tumors of lung and bronchus:
1. Primary
2. secondary

Diseases of PLEURA:

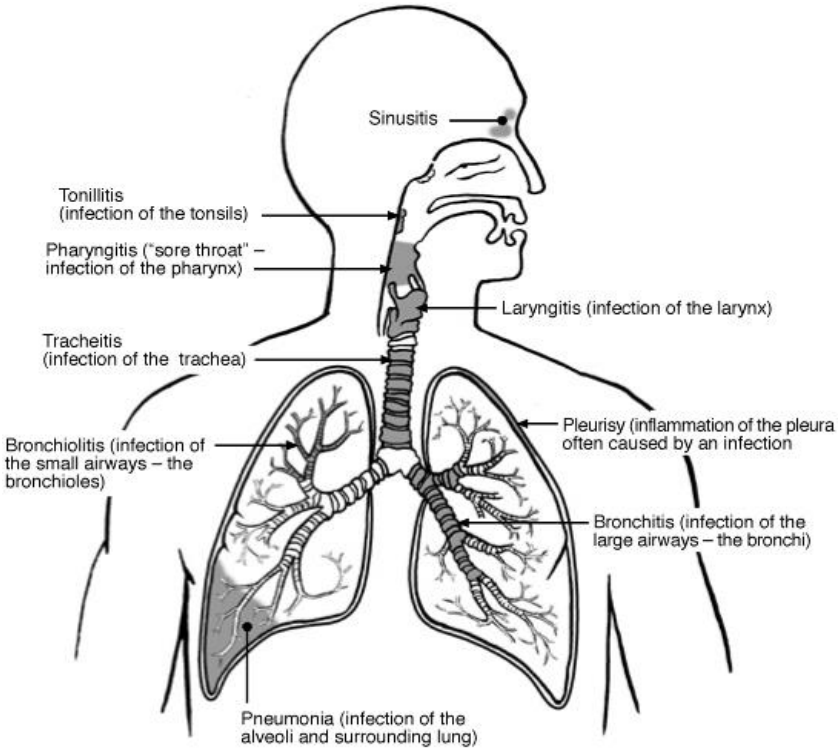
Pleurisy: The term used for any diseases involving pleura and giving rise to pleural pain or pleural friction.

Pleural effusion:
Accumulation of fluid in the pleural space.
Causes:
-Pneumonia
-TB
-Pulmonary infarction
-Cardiac failure

Empyema: Pus in the pleural space

Pneumothorax: Air in the pleural space

Haemothorax: Blood in the pleural space



Infections of the respiratory tract

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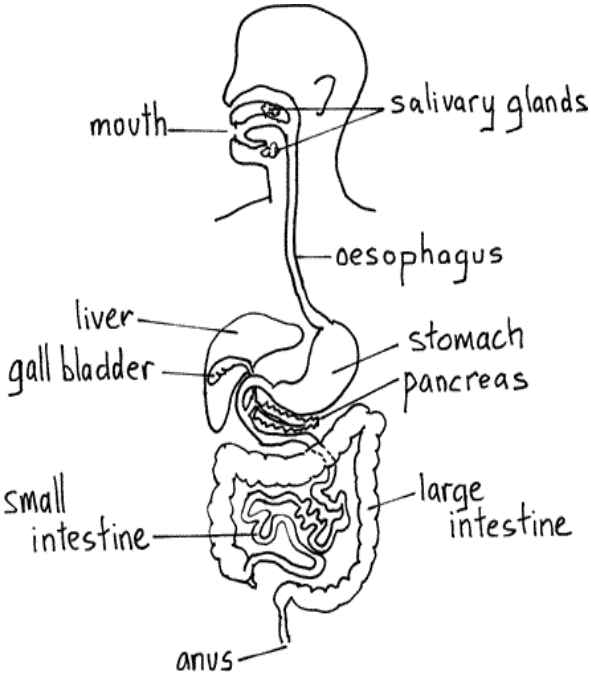
Chapter 4
Gastro Intestinal System & Hepato-Biliary System:

GASTRO INTESTINAL SYSTEM

Anatomy:



It consists of the following organs:

- Mouth
- Pharynx
- Esophagus
- Stomach
- Pancreas
- Small intestine:
    1. Duodenum
    2. Jejunum
    3. Ileum
- Large Intestine:
  Colon- Ascending, Transverse, Descending
- Rectum
- Anus

Functions of the alimentary tract:

- Ingesting and absorbing nutrients
- Excreting unabsorbed and waste products

Individual Functions

ESOPHAGUS

- Propulsion of food from mouth to stomach

STOMACH

- Digestion of food

SMALL INTESTINE

- Further digestion and absorption

LARGE INTESTINE

- Absorption and excretion

DISEASES

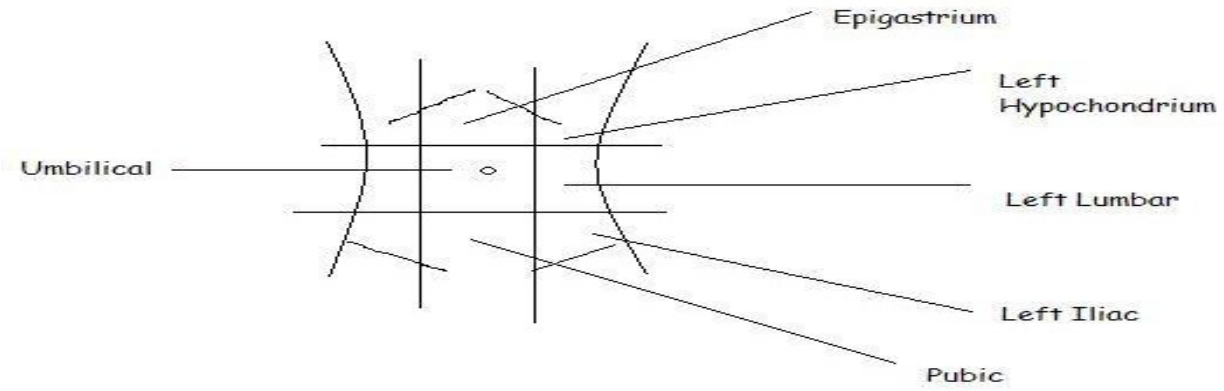
MOUTH ULCER

- They are red, painful, superficial ulcers in the mouth.
- Associated with vitamin deficiency, Stress, Crohn's Disease and Ulcerative Colitis.

LEUKOPLAKIA

- It is a painless white lesion. It appears as a white, smooth firm patch mostly on the sides of the tongue.
- It is a precancerous lesion

CLINICAL REGIONS OF ABDOMEN



GASTRO-ESOPHAGEAL REFLUX DISEASE (GERD)

There is a sphincter located between the stomach and esophagus. This prevents the gastric acid entering the esophagus.
In GERD, the sphincter is weakened and the acid and food from the stomach passes into the esophagus. The esophageal lining is damaged due to the effect of the acid.

Causes:

- Stress
- Obesity
- Smoking
- Psychological
- Irregular food habits- over eating
- Lack of exercise
- Pregnancy

Symptoms:

- Heart Burn
- Regurgitation of Gastric content in the mouth.

Cancer of the Esophagus can develop in long standing cases of GERD.

DYSPHAGIA: Difficulty in swallowing
Causes:

- Mouth:
  - -Stomatitis
  - -Mouth Ulcers
  - -Malignancy
- Esophagus
  - -Loss of motility due to any cause
  - -Motor Neuron Disease
  - -Malignancy
  - -GERD

GASTRITIS
Causes:

- NSAIDs
- Alcohol
- Drugs
- Stress
- Infection- H. pylori

ACID PEPTIC DISEASE (APD) – PEPTIC ULCER
Causes:

- Chronic Stress
- Alcohol
- Drugs
- Smoking
- H. Pylori infection

Clinical Features:

- Epigastrric Pain
- Night Pain
- Hunger Pain

Complications:

- Perforation
- Peritonitis
- Cancer

***Peritonium is a transparent double layered membrane covering all the organs of the abdomen and pelvis.**

PERITONITIS

- Inflammation of the peritoneum.

ASCITES

- Accumulation of fluid in the peritoneal cavity

INTESTINAL OBSTRUCTION

Causes:

- Worms
- Faecal Impaction
- Tumours
- Volvulus/ Intussusception
- Strictures

Symptoms:

- Acute pain
- No peristaltic movement

PANCREATITS

Causes:

- Alcohol
- Gall stones
- Anatomical Obstruction
- Drugs
- Infections (Mumps)

Clinical Features:

- Acute pain in the epigastrium or right hypochondrium, often following heavy alcohol consumption

ULCERATIVE COLITIS & CHRON'S DISEASE

- CD- throughout the alimentary tract and entire thickness of the wall involved;
  UC- only large intestine involved, only mucosa layer involved
- CD- Cobblestone appearance
  UC- Haemorrhagic ulcer

Cause:

- Genetic
- Infective agents
- Smoking

DIARROHEA

More than 3 bowel movements in a day or semi liquid stool formation irrespective of the number of the motions.

Causes:
- Infection
- Intolerance
- Motility Disorder
- Drug ADR (Adverse Drug Reaction)

CONSTIPATION

Less than 3 bowel movements in a week
Causes
- Irregular food habits
- Less dietary fibre intake
- Drug ADR
- Decreased fluid intake

IRRITABLE BOWEL SYNDROME

Prolonged constipation or Diarrhoea, alternating or alone, with dull pain in the epigastrium

Causes
- Food Intolerance
- Stress
- Autoimmune

HAEMORRHOIDS (PILES)

Enlarged veins protruding into the rectum or into the anus
Causes:
- Prolonged constipation

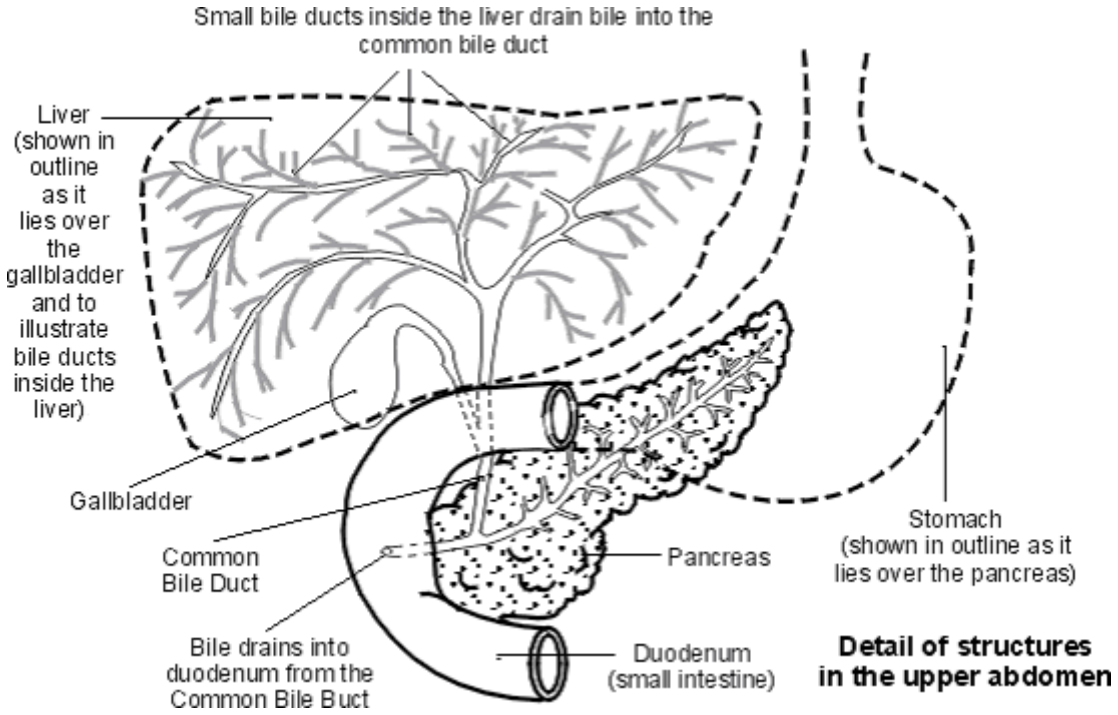
Clinical Features:
- Blood in stools
- Pain during defecation

FISSURES

Tear in the anus.
Causes:
- Prolonged constipation

Clinical Features:
- Blood in stools
- Pain during defecation

HEPATO-BILIARY SYSTEM



COMPONENTS OF HEPATO-BILIARY SYSTEM

1. Liver
2. Gall Bladder
3. Biliary Duct

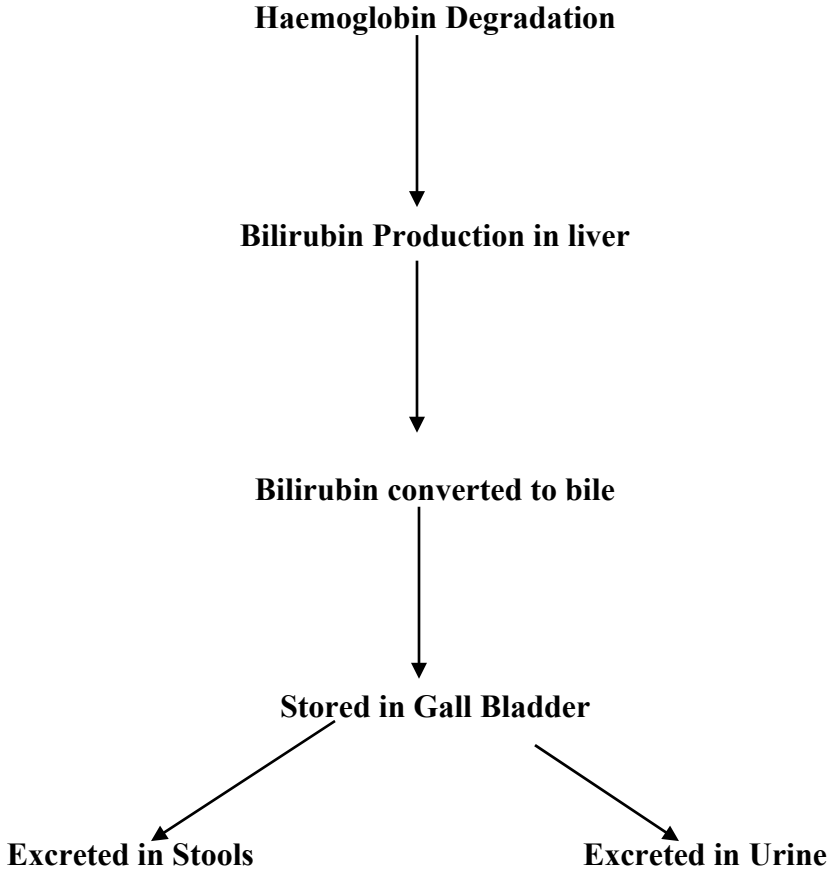
FUNCTIONS OF LIVER:

- Metabolism- proteins, Fats, Carbohydrates
- Metabolism of haem
- Drug Metabolism
- Detoxification of drugs and poisons
- Degradation of alcohol
- Bile formation

FUNCTIONS OF GALL BLADDER

- Storing and secretion of bile

BILIRUBIN METABOLISM



JAUNDICE
Yellow discolouration of skin and mucous membrane
Types:
1. Haemolytic
2. Hepatocellular
3. Obstructive

HEPATITIS

Causes
- Infection- Hepatits A, Hepatitis B,
- Drugs
- Alcohol
- Poison

LIVER CIRRHOSIS
Death of Liver cells

OBSTRUCTIVE JAUNDICE

Causes:

- Gall stones
- Strictures
- Cancer

LIVER ABSCESS

Causes:

- Amoebic liver abscess
- Pyogenic liver abscess

AMOEBIC LIVER ABSCESS

- Cause – E. histolytica
- Transmitted by feco-oral route

Pathology:

Protozoa enters through contaminated food and water → Colonizes in the large intestine normally causing dysentery → Sometimes the protozoa colonize the liver tissue forming an cyst

*Cyst- is an accumulation of fluid/pus >1cm in diameter surrounded by a membrane

GALL STONES

Clinical Features:

- Pain in the right hypochondriac region
- Jaundice

Complications

- Cholecystitis- inflammation of gall bladder
- Hepatitis
- Cancer

Chapter 5

Endocrine Glands

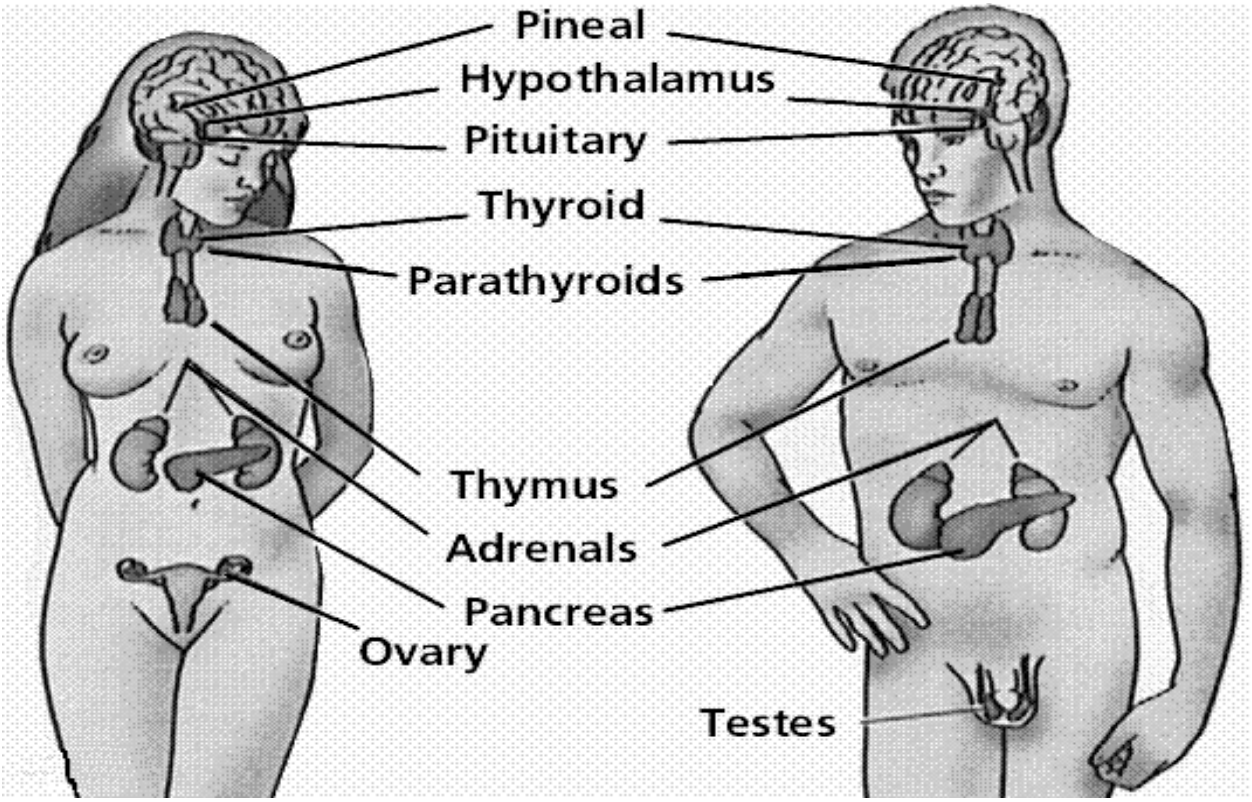
- Endocrine glands are specialized cluster of cells that secrete hormones.

  locally – paracrine .Distant – endocrine.Secreted hormones go directly into the blood stream (ductless gland) in response to the nervous system stimulation.
- Integrate functions of physiologic systems.
- Consists of several glands located in various parts of the body.
- Endocrine glands include **:**

1 Hypothalamus ,1 pituitary gland, 1thyroid gland,4 parathyroid glands, 2 adrenal glands, 1thymus Pineal gland, 2 ovaries and testes

- Controls many body functions

Exerts control by releasing special chemical substances into the blood called ***hormones.*** ***Hormones*** affect other endocrine glands or body systems. Secreted in minimum amount in response to need. Either travel through the blood stream to the target organ or are secreted locally to produce an effect.
Transportation of the hormones - Bounded to plasma proteins such thyroid and steroid (they serve as a reserve for acute changes). Some are transported free in the blood only free hormones are biological active. Maintain homeostatic balance utilizing a feedback mechanism that involves other hormones, blood or chemicals, and the nervous system.When the sensor detects a decreased in hormone levels. They began to act to cause at increased in hormonal level. When the hormonal levels rise above normal, the sensors cause a decreased in hormonal production. Sensors in the endocrine system detect changes in the hormonal levels.
Hormones are adjusted to maintain normal body levels.



Hypothalamus and Anterior Pituitary –

- **Pituitary gland**: a small gland located on a stalk hanging from the base of the brain –

"**The Master Gland"** Primary function is to control other glands. Produces many hormones. Secretion is controlled by the hypothalamus in the base of the brain.

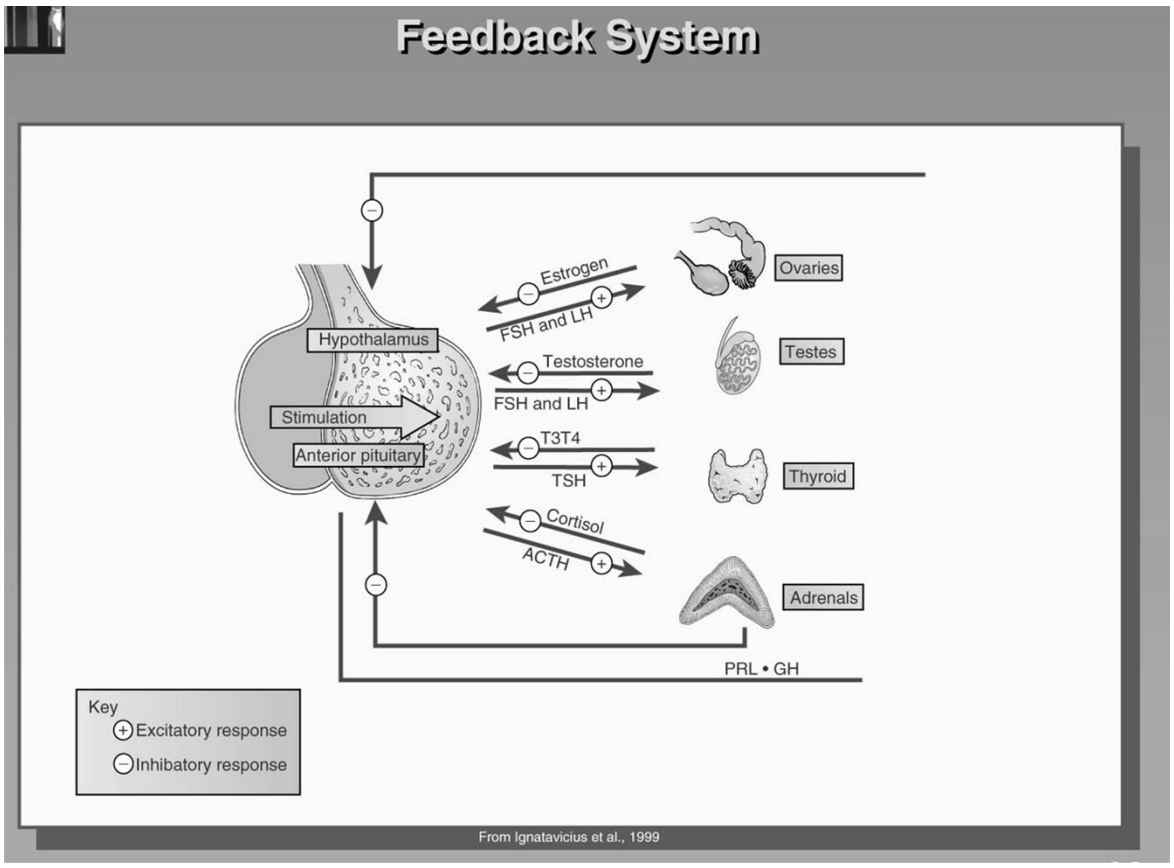
- The Pituitary Gland is divided into two areas, which differ *structurally* and *functionally*

Each area has separate types of hormone production.

Anterior Lobe produces seven Hormones:
ACTH – adrenocorticotrophic hormone, GH – growth hormone – also known as somatotrophin, PRL - prolactin, FSH – follicle stimulating hormone, LH – luteninzing hormone- female / ICSH – interstitial cell stimulating hormone – male, TSH – thyroid stimulating hormone MSH – melanocyte stimulating hormone

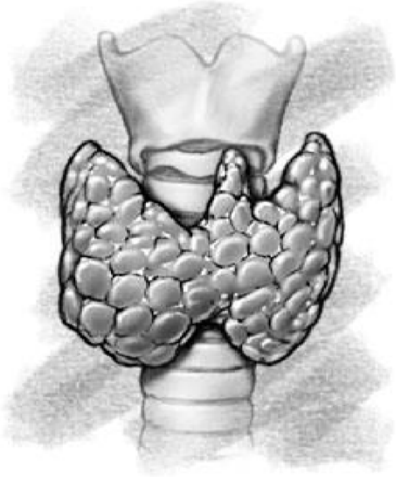
Posterior Lobe Produces oxytocin , and antidiuretic hormone (ADH)

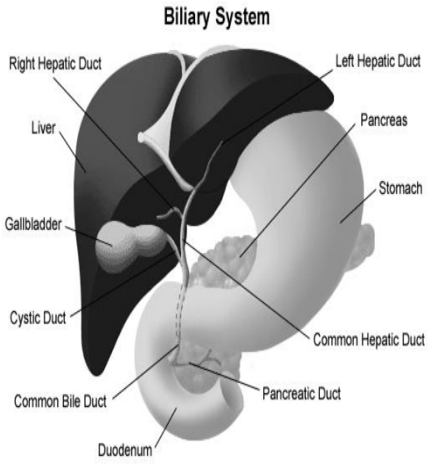
Thyroid gland A small gland shaped like a butterfly located below the larynx; it weights 15-20g. It consists of two lobes, one on either side of the thyroid cartilage.Needs iodine to produce hormones It produces these two hormone thyroxine (T4) and triiodothyronine (T3)
TSH – thyroid stimulating hormone - Causes production of thyroid hormones [thyroxine T_4 and triiodothyronine T_3] that maintain metabolism. Stimulation: Stress, malnutrition, low plasma glucose and sleep Regulation: Negative feedback control.First, the hypothalamus, located just above the pituitary gland in the brain, secretes Thyrotropin which causes the pituitary gland to produce thyroid-stimulating hormone (TSH). Just as the name suggests, TSH stimulates the thyroid gland to produce thyroid hormones.The pituitary gland slows or speeds the release of TSH, depending on whether the levels of thyroid hormones circulating in the blood are getting too high or too low

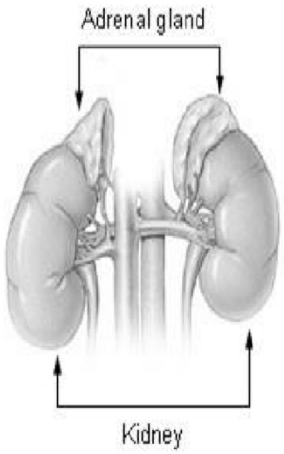


Functions of thyroid hormones. They act on most body systems usually stimulating them Metabolism : Controls and increased the basic metabolic rate (BMR) increasing oxygen consumption and heat production.Carbohydrate metabolism: Stimulates cellular glucose uptake, Glycolysis, Gluconeogenesis, GI absorption and insulin secretion.Body Tissues Metabolism – BMR .Thyrocalcitonin Bone Calcium ++ ,CHO Metabolism ,Fat Metabolism, Protein Metabolism, Body Weight, Body Growth, Cardiovascular Function, Gastrointestinal function , CNS

Parathyroid Glands - small, pea-shaped glands, located in the neck near the thyroid.Usually 4 - number can vary . Regulate the level of calcium in the body. Produce ***parathyroid hormone -*** ↑level of calcium in blood . Hypocalcemia can result if parathyroids are removed or destroyed.







Thyroid Gland
Pancreas and its surrounding structures Adrenals
Pancreas a key gland located in the folds of the duodenum . Has both endocrine and exocrine Functions . Secretes several key digestive enzymes. Islets of Langerhans -specialized tissues in which the endocrine functions of the pancreas occurs . include 3 types of cells:

- alpha (α)
- beta (β)
- delta (∂)

each secretes an important hormone. **Alpha (α) cells** release *glucagon*, essential for controlling blood glucose levels. When blood glucose levels fall, αcells ↑ the amount of glucagon in the blood.The surge of glucagon stimulates the liver to release glucose stores (from glycogen and additional storage sites).Also, glucagon stimulates the liver to manufacture glucose – gluconeogenesis.Beta Cells release insulin (antagonistic to glucagon). Insulin the rate at which various body cells take up glucose. Thus, insulin lowers the blood glucose level.

Insulin is rapidly broken down by the liver and must be secreted constantly.Delta Cells produce somatostatin, which inhibits both glucagon and insulin.Disorder of the pancreatic islets leads to Diabetes mellitus which leads to type1 insulin dependent diabetes mellitus or type 2 non-insulin dependent diabetes mellitus.

Type 1: insulin dependent diabetes mellitus.Occurs mainly in young children. Juvenile onset diabetes mellitus.Deficiency of insulin is due to the destruction of islet cells. Less common .Low degree of genetic predisposition . The onset is usually sudden.

Type 2: non insulin dependent diabetes mellitus .Maturity onset diabetes mellitus. Level of insulin is low. Most common. Has high degree of genetic predisposition.Has a late onset . It is characterized by hyperglycemia,polydipsia,polyuria,weight loss.

Adrenal Glands 2 small glands that sit atop both kidneys. Each has 2 divisions, each with different functions.The Adrenal Medulla secretes the catecholamine hormones nor-epinephrine and epinephrine (closely related to the sympathetic component of the autonomic nervous system).The Adrenal Cortex secretes three classes of hormones – all steroid hormones: gluticocorticoids , mineralocorticoids, androgenic hormones.

Gluticocorticoids: accounts for 95% of adrenal cortex hormone production. Increase the level glucose in the blood. Released in response to stress, injury, or serious infection - like the hormones from the adrenal medulla.

Mineralocorticoids: Regulate the concentration of potassium and sodium in the body. Prolonged increase in adrenal cortex hormone results in Cushing's Disease .

Cushing's Syndrome : Caused by excess of cortisol production or by excessive use of cortisol or other similar steroid (glucorticoid)

Addison's Disease : Addison's disease is a severe or total deficiency of the hormones made in the adrenal cortex, cause by a destruction of the adrenal cortex.Prolonged ↑ in adrenal cortex hormone results in *Cushing's Disease. Signs and Symptoms of Cushings Disease* –Increase in blood sugar levels , unusual body fat distribution , rapid mood swings, Immunosupression ,

Osteoporosis , Emotional liability, Hypokalemia , Hypertension, changes in body appearance, Weakness, Fatigue, Increased thirst and urination, Lack of menstrual periods
Addison's Disease**:** It is a hypofunction of the adrenal cortex.Adrenal glands do not produce enough of the adrenal cortisol. Rare disorder can occur at any age; most common among adult white women. **Signs and symptons of addison's -** Chronic fatigue , Muscle weakness, Loss of appetite, Weight loss, Nausea/vomiting, Low blood Pressure , Hyperpigmentation, Irritability/depression.And - if there is an ↑in mineralocorticoids as well, a serious electolyte imbalance will occur due to the ↑potassium excretion by the kidney, which results in hypokalemia.

Testes And Ovaries the endocrine glands associated with human reproduction. Female ovaries produce eggs .Male gonads produce sperm . Both have endocrine functions.

Ovaries - located in the abdominal cavity adjacent to the uterus. Under the control of LH and FSH from the anterior pituitary they manufacture estrogen and protesterone. Estrogen and Progesterone have several functions, including sexual development and preparation of the uterus for implantation of the egg

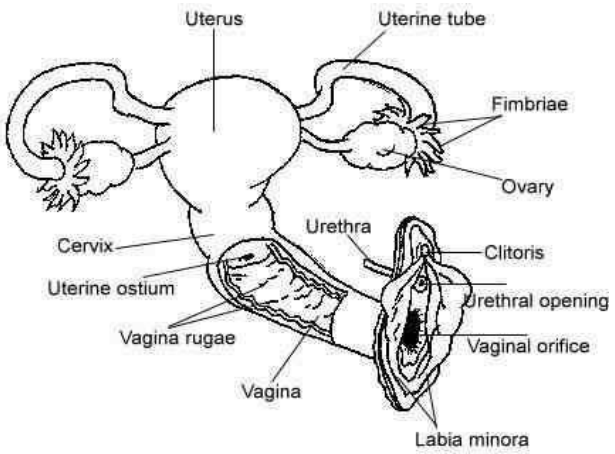
Testes - located in the scrotum. Produce sperm for reproduction. Manufacture testosterone Promotes male growth and masculinization. Controlled by anterior pituitary hormones FSH and LH.

Chapter 6
The Reproductive System

Components -
Gonads: organs that produce gametes and hormones
Ducts: receive and transport gametes
Accessory glands: secrete fluids into ducts
Perineal structures: collectively known as external genitalia.
Male & Female Reproductive Systems Are functionally different . Female produces 1 gamete . per month, retains and nurtures zygote . Male disseminates large quantities of gametes, produces 1/2 billion sperm per day.

Female Reproductive System

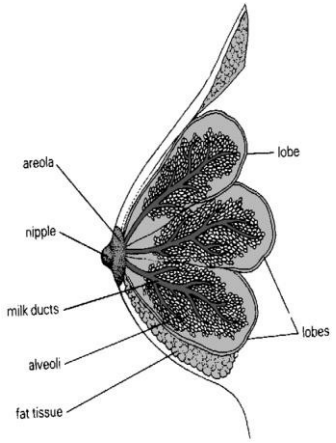
Female Reproductive System **-** Produce sex hormones. Produce functioning gamates [ova] . Support & protect developing embryo



Vagina:
Passageway between the uterus and the outside of a woman's body.
Cervix:
Opening from the uterus to the vagina.
Uterus:
Place where the baby grows in a woman's abdomen.

Uterine Tubes (Fallopian Tubes):
Two tubular structures leading from the ovaries to the uterus
Ovaries:
organs holding a woman's eggs.

Mammary Glands[breasts] - Present in both sexes - normally only functional in females.



Developmentally they are derived from sweat glands. Slightly below center of each breast is a ring of pigmented skin, the areola - this surrounds a central protruding nipple. Internally - they consist of 15 to 25 lobes that radiate around and open at the nipple. Each lobe is composed of smaller lobules- these contain alveoli that produce milk when a women is lactating. Non-pregnant women - glandular structure is undeveloped - hence breast size is largely due to the amount of fat deposits.

The Menstrual cycle - Is th Menstural phase

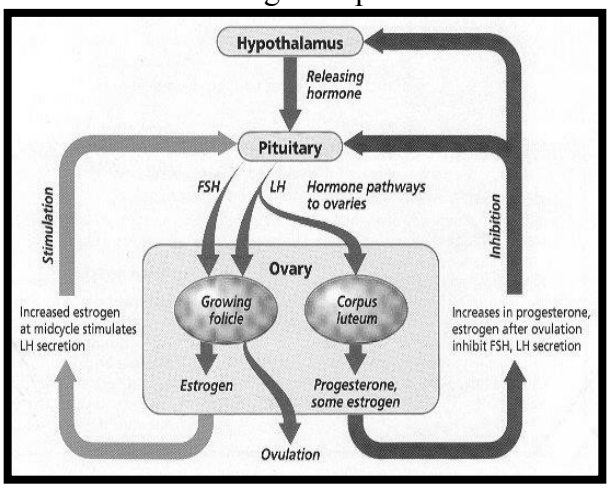
Proliferative phase Secretory phase

Menstrual Phase : The first day of the menstrual period is considered Day 1 of the cycle.During this time, the endometrium (the built-up lining of the uterus) is shed, along with a little blood.Many of the problems that women experience with their menstrual cycle occur during this phase.For example, some women experience menstrual disorders such as dysmenorrhea (painful periods) or menorrhagia (unusually heavy periods).

The Follicular/Proliferate Phase : During the proliferate phase, the body produces a hormone called follicle-stimulating hormone (FSH).Follicle-stimulating hormone promotes the growth of a follicle (egg sac) within the ovary.An ovum (egg) matures in the follicle during the proliferative phase.FSH also stimulates the ovary to produce increasing amounts of estrogen.In turn, the estrogen causes endometrial tissue to build up (or proliferate), lining the interior of the uterus.

The Luteal/Secretory Phase :
Once the ovum has been released, the follicle becomes a sac known as the corpus luteum ("yellow body," because it contains yellowish, fatty matter).

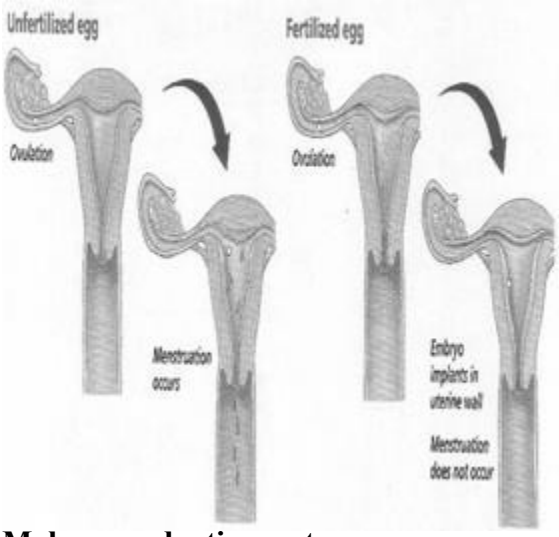
A hormone called luteinizing hormone (LH) causes the corpus luteum to grow and to secrete progesterone, another female hormone. During the secretory phase, progesterone makes the endometrial lining stronger and spongy in texture.Progesterone also stimulates glands in the endometrium.These glands produce uterine fluid, and their purpose is to support embryonic development if fertilization has occurred at or around the time of ovulation.It is in this phase of the menstrual cycle that women who suffer from premenstrual syndrome (PMS) may begin to experience their symptoms.The flow of blood to the endometrium decreases, and its upper portion is broken down and shed during menstruation.



At the same time, the corpus luteum withers.

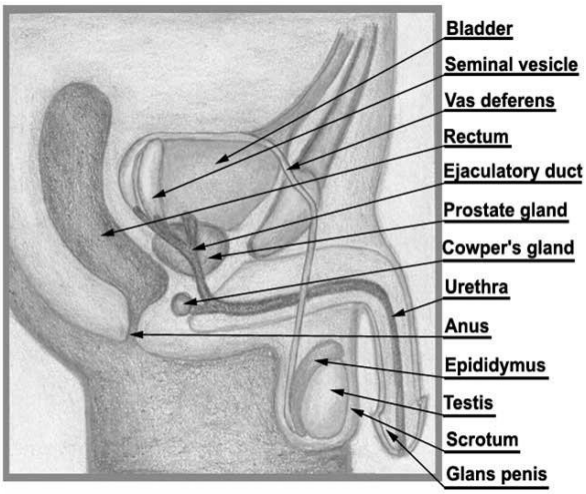
When does ovulation occur? The timing of ovulation varies with the length of a woman's menstrual cycle. In the average 28 day menstrual cycle, the LH surge usually occurs between cycle days 11-13 and ovulation follows about 36-48 hours later, on or close to cycle day 14. Women with shorter menstrual cycle lengths tend to ovulate earlier and women with longer cycle lengths tend to ovulate later than cycle day 14. Despite the variations in menstrual cycle length, the time from ovulation to the onset of the next menstrual period is usually constant (2 weeks). This principle is the basis for the use of ovulation calendars that take into account an individual's shortest and longest cycle lengths.

Fertilization (conception): A sperm entering an ovum. Follicle-stimulating Hormone (FHS): a



substance which brings to life a few of the ovum in one of the ovaries.
Luteinizing Hormone (LH): causes the follicle to burst, and allows ovum to fall into the opening of the fallopian tube.
Ova –plural, Ovum—singular: the female reproductive cell.
Ovaries: organs holding a woman's eggs.

Male reproductive system :
What are the components of the male reproductive system?
The Components consist of
Testes
Epididymides
Deferent ducts
Spermatic cords
Seminal vesicle
Ejaculatory ducts
Prostate gland
Penis
External Male Reproductive System :
Testosterone: The male reproductive hormone made by the testicles which causes the changes of puberty.



Penis: the organ of transfer of sperm to female.
Scrotum:pouch-like sac holding both testicles in a separate compartment that hang underneath the penis.
Testicles– Testes Gland: Two glands in the male, located in the scrotum, which produce male hormones (testosterone).

Epididymis : A long tube (about 6 meters) located along the superior and posterior margins of the testes. Sperm that leave the testes are immature and incapable of fertilizing ova. They complete their maturation process and become fertile as they move through the epididymis. Mature sperm are stored in the lower portion, or tail, of the epididymis. There are two spermatic cords,one leading from each testis consisting proximal ductus deferens, testicular artery and veins, lymph vessels, testicular nerve, and a connective tissue covering.

Duct System : Sperm cells pass through a series of ducts to reach the outside of the body. After they leave the testes, the sperm passes through the epididymis, ductus deferens, ejaculatory duct, and urethra.

Seminal Vesicles - Two pouch-like structures which serve to store mature sperm until ejaculated.

Internal Male Organs - Sperm: the microscopic cells produced by the male's testicles which can fertilize the female's ovum.

Prostate Gland: a man's gland that helps make semen.

Cowper's Glands: behind the base of the penis which secretes fluid to make semen and neutralize acid during sexual excitement.

Ejaculatory duct: a short straight tube that passes into the prostate gland and opens into the urethra.

Erection : Involves increase in length, width & firmness.Changes in blood supply: arterioles dilate, veins constrict.The spongy erectile tissue fills with blood .Erectile Dysfunction [ED] also known as impotence.

Hormones : Follicle-stimulating hormone (FSH) stimulates spermatogenesis. Interstitial Cell Stimulating Hormone (ICSH) stimulates the production of testosterone . Testosterone stimulates the development of male secondary sex characteristics & spermatogenesis.

Prostate Cancer : Common in men older than 50; ranks high as cause of cancer death. Leading cause of death from cancer .
Signs and Symptoms : Hard nodule in periphery of gland. Detected by rectal exam. As tumor develops, obstruction occurs . Hesitancy, decreased stream, urinary frequency, bladder infection

Benign Prostatic Hypertrophy : Common in older men; varies from mild to severe. Change is actually hyperplasia of prostate.Nodules form around urethra.Result of imbalance between estrogen and testosterone. No connection with prostate cancer. Rectal exams reveals enlarged gland. Incomplete emptying of bladder leads to infections. Continued obstruction leads to distended bladder, dilated ureters, renal damage.

Chapter 7
The Nervous System

A network of billions of nerve cells linked together in a highly organized fashion to form the rapid control center of the body.

Basic Functions of the Nervous System :

Sensation

Monitors changes/events occurring in and outside the body. Such changes are known as stimuli and the cells that monitor them are receptors.

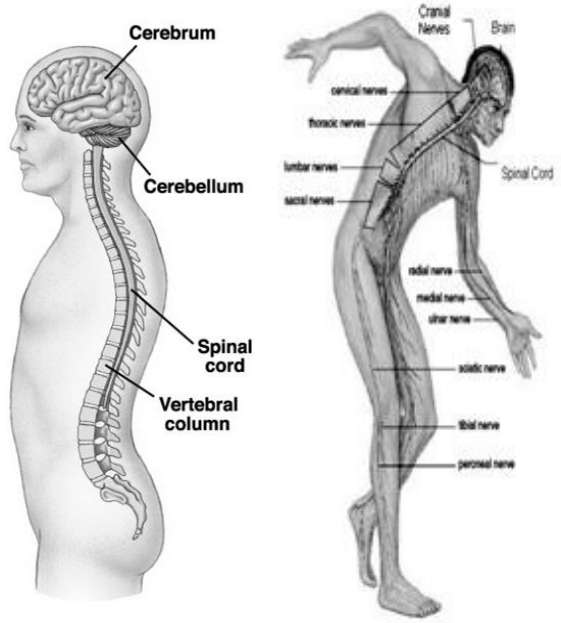
Integration

The parallel processing and interpretation of sensory information to determine the appropriate response

Reaction

Motor output. The activation of muscles or glands (typically via the release of neurotransmitters (NTs)

Organization of the Nervous System



Nervous System is divided into:

Central Nervous System - The brain and the spinal cord , the center of integration and control.

Peripheral Nervous System - The nervous system outside of the brain and spinal cord. Consists of:

31 Spinal nerves, carry info to and from the spinal cord.

12 Cranial nerves , carry info to and from the brain.

Central Nervous System :

We attribute intellect, as well a host of other functions to the brain. That grayish matter resting with in

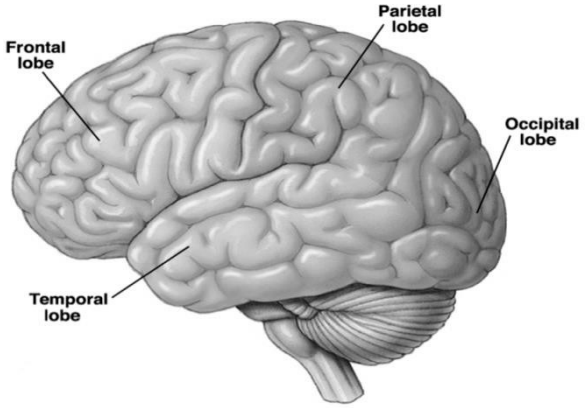
the bony cranium. Weighs about 1600g in males and about 1400g in females .Has about 10^{12} neurons,

each of which may receive as many as 200,000 synapses . Although these numbers connote a high level of complexity, the CNS is actually quite orderly.

Gray and White Matter - Viewed macroscopically, CNS tissues can be distinguished by color.

Brain Regions – Cerebrum, Diencephalon, Brainstem and Cerebellum

Cerebrum - The largest, most conspicuous portion of the brain.Two hemispheres connected by the
corpus callosum.Has an outer cortex of gray matter surrounding an interior that is mostly white matter, except for a few small portions.The surface is marked by ridges called gyri separated by grooves called sulci. Deep sulci divide each hemisphere into lobes.



- 3 types of functional areas:

Motor → Control voluntary motor functions
Sensory → Allow for conscious recognition of stimuli
Association → Integration

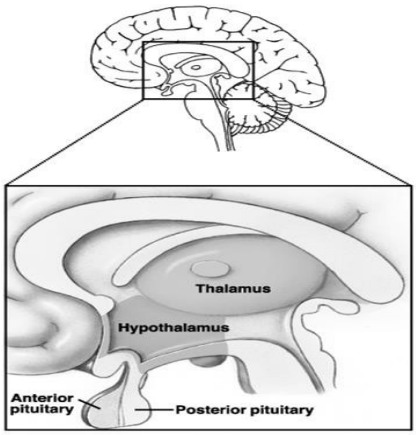
Prefrontal Cortex - Anterior frontal lobes.
Involved in analysis, cognition, thinking, personality, conscience.

The Left side of our brain functions in a different manner than the right.Certain activities are the almost exclusive domain of one of the two hemispheres.The left side of our brain perceives the world in a linear manner.It tends to organize sensory inputs into the form of points on a line.For example, language which is linear is a function of the Left Hemisphere. The Left Hemisphere functions logically and rationally. It is the left side of the brain which creates the concept of causality, the image that one thing causes another because it always precedes it.The right hemisphere by comparison, perceives whole patterns.In most people, the left hemisphere has a more control over language, math, and logic.While the right hemisphere is geared towards musical, artistic and other creative endeavors.Physiologically, the right hemisphere controls the left side of the body.Most individuals with left cerebral dominance are right-handed.

Basal Nuclei - Includes theCaudate Nucleus,Lentiform Nucleus which includes Globus pallidus and Putamen. Components of the extrapyramidal system which provides subconscious control of skeletal muscle tone and coordinates learned movement patterns and other somatic motor activities.The caudate nucleus is especially affected in Huntington's chorea. Other components of the extrapyramidal system : Substantia nigra,Parts of the thalamus,Subthalamic nucleus, Red NucleusThe Substantia nigra, is located in the midbrain. It is particularly affected in Parkinson's disease.

Diencephalon Forms the central core of the forebrain.3 paired structures:Thalamus, Hypothalamus , Epithalamus

Thalamus - 80% of the diencephalon Sensory relay station where sensory signals



and routed.Also has profound input on motor (via the basal ganglia and cerebellum) and cognitivefunction.

HypothalamusFunctions:

Autonomic regulatory center - Influences HR, BP, resp. rate, GI motility, pupillary diameter.

Emotional response

Involved in fear, loathing, pleasureDrive center: sex, hunger can be edited, sorted,

Regulation of body temperature, Regulation of food intake ,Regulation of water , balance and thirst

Regulation of sleep/wake cycles , Hormonal control .

Epithalamus Contains the pineal gland which releases melatonin (involved in sleep/wake cycle and mood). Contains a structure called the habenula – involved in food and water intake.

Brain Stem : Consists of Mid Brain, Pons , Medulla Oblongata

Cerebellum : Lies inferior to the cerebrum and occupies the posterior cranial fossa.Second largest region of the brain. 10% of the brain by volume, but it contains 50% of its neurons. Has two primary functions:

1. Adjusting the postural muscles of the body- Coordinates rapid, automatic adjustments, that maintain balance and equilibrium.
2. Programming and fine-tuning movements controlled at the subconscious and conscious levels

The cerebellum can be permanently damaged by trauma or stroke or temporarily affected by drugs such as alcohol.These alterations can produce ataxia – a disturbance in balance.

Blood Supply to the Brain : Disruption in blood supply results in Stroke

Nervous Tissue : 2 cell types – Neurons - Functional, signal conducting cells

- Neuroglia - Supporting cells

Defective conduction leads to epilepsy.

Peripheral Nervous System

31 spinal nerves

12 cranial nerves

Cranial Nerves

Olfactory,Optic, Oculomotor ,Trochlear ,Trigeminal, Abducens , Facial ,Vestibulocochlear , Glossopharyngeal ,Vagus, Accessory, Hypoglossal

The 31 pairs of spinal nerves are grouped as follows: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal.

Autonomic Nervous System :

Two divisions: Sympathetic and Parasympathetic

Sympathetic - "Fight or flight"

"E" division - Exercise, excitement, emergency, and embarrassment

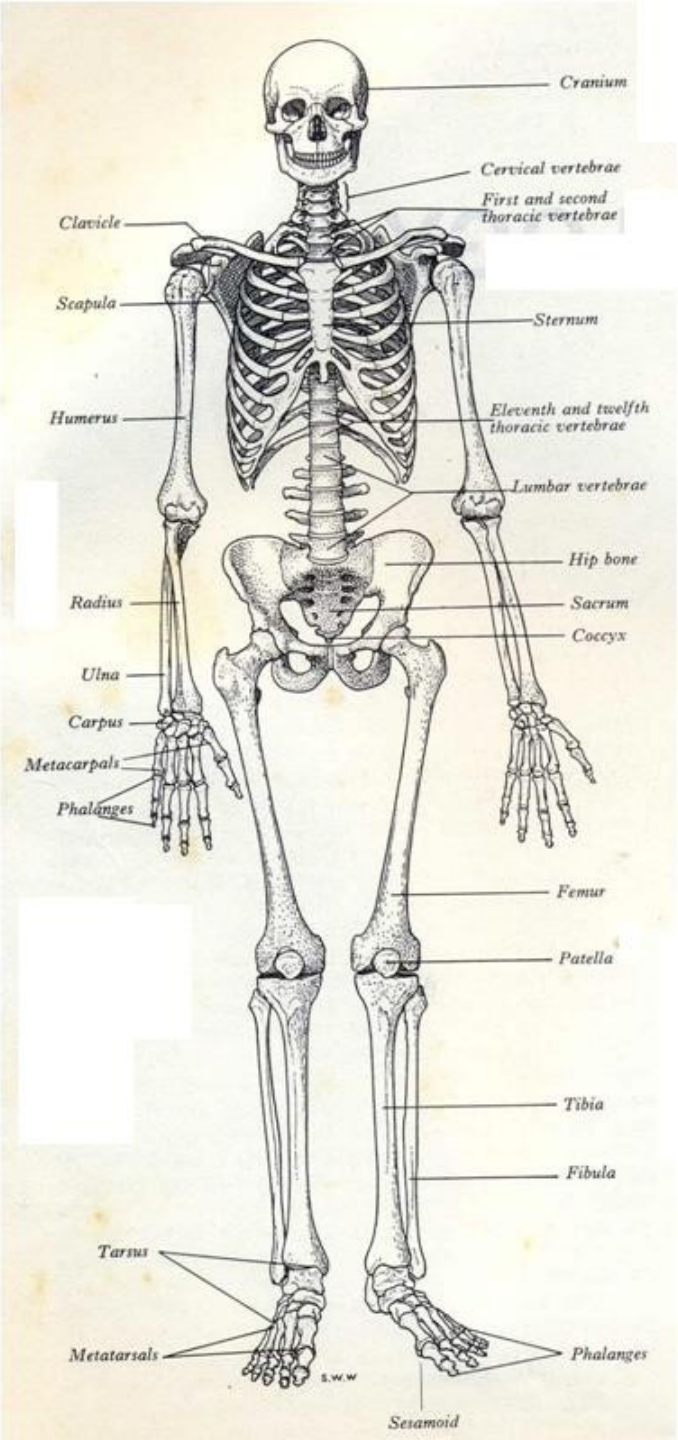
Parasympathetic -"Rest and digest"

"D" division - Digestion, defecation, and dieresis

How Does the Brain Control the ANS? - The hypothalamus is the Boss.Its anterior and medial regions direct parasympathetic function while its posterior and lateral regions direct sympathetic function.These centers exert control directly and via nuclei in the reticular formation (e.g., the cardiovascular centers in the Medula Oblongata, respiratory centers in Medula Oblongata and pons, etc.)The connection of the limbic system to the hypothalamus mediates our "fight or flight" response to emotional situations.The relationship between the hypothalamus and the amygdala allows us to respond to fear.

Somatic nervous system The somatic nervous system is the part of the peripheral nervous system associated with the voluntary control of body movements through the action of skeletal muscles, and with reception of external stimuli, which helps keep the body in touch with its surroundings (e.g., touch, hearing, and sight).The system includes all the neurons connected with muscles, skin and sense organs.

Chapter 8
Skeletal System



Spinal column

7-Cervical-1st-Atlas
-2nd –Axis(pivot on which atlas ,hence head rotates)
12-Thoracic
5-Lumbar
5-Sacral bones-all fused together
3-5—rudimentary bones of coccyx

Sternum - probably confined to land animals. Absent in fish.
Clavicle - The bone that helps transmit part of weight of upper arm to axial skeleton.
Humerus - longest and largest bone of upper limb.
Radius - Lateral bone of arm broader below.
Ulna - Medial bone of arm runs parallel to radius.
Wrist – Carpal bones (wrist)
- Metacarpal bones of palm
-Phalanges or bones of digits.
Carpal bones-8 bones-scaphoid
-lunate
-triquetral
-pisiform
-trapezium
-trapezoid
-capitate
-hamate
Metacarpals-5
Phalenges-14-3 in each finger
-2 in thumb
Hip bone – ilium
-ischium
-pubis.
Femur-thigh bone-largest and strongest bone.
Patella –largest sesmoid bone in front of knee joint.
Tibia- medial-much stronger of the two bones of the leg.
Fibula –lateral-slender, not called upon to share in transmission of body weight.
Foot –tarsus-7 bones (like wrist but bear & distribute weight)
-(talus,calcaneus,medial cuneform,intermediate cuneiform, lateral cuneiform, cuboid, navicular)
-metatarsus-5
-phalanges-same as hand